

Cisco Firepower Next-Generation Firewall VPN Lab – FDM v1.6

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About This Guide

This guide for the preconfigured demonstration includes:

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Requirements

The table below outlines the requirements for this preconfigured demonstration.

Table 1. Requirements

Required	Optional
Laptop	Cisco AnyConnect®

About this Solution

IT teams have been asked to manage security using a patchwork of siloed point products, starting with legacy next-generation firewalls (NGFW), which were created with a focus on applications and bolted on best effort threat protection. As such, these legacy NGFWs are unable to provide an enterprise with the contextual information, automation, and prioritization required to handle today's modern threats.

Cisco Firepower is an integrated suite of network security and traffic management products, deployed either on purpose-built platforms or as a software solution. The system is designed to help you handle network traffic in a way that complies with your organization's security policy - your guidelines for protecting your network.

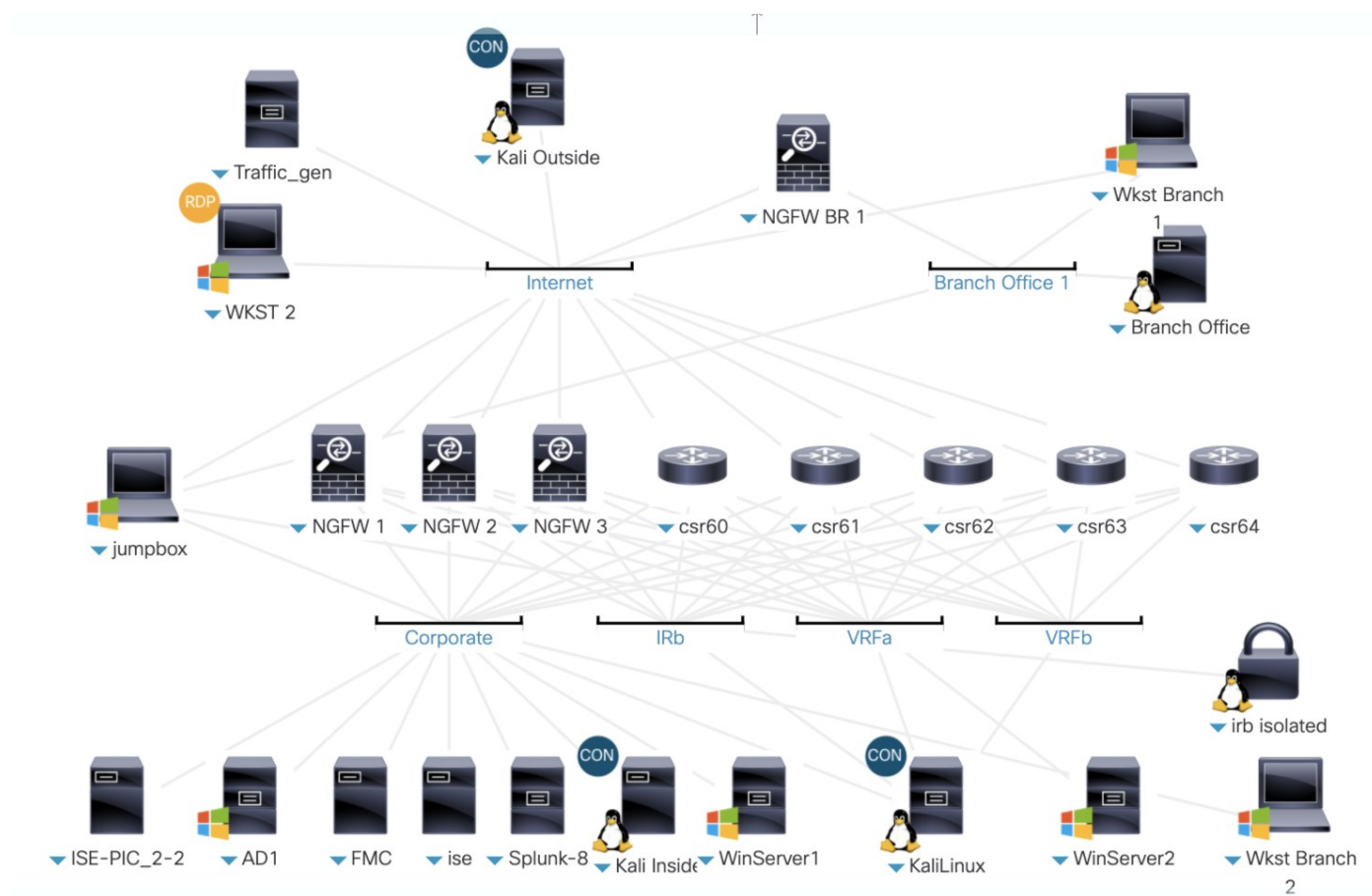
This allows the Cisco Firepower NGFW to evolve with a focus on enabling enterprises to stop, prioritize, understand, and automate responses to modern threats in real-time. Firepower NGFW is unique in its threat-focus, with a foundation of comprehensive network visibility, best-of-breed threat intelligence and highly effective threat prevention to address both known and unknown threats. Firepower NGFW also enables retrospective security, through Advanced Malware Protection, that can "go back in time" to quickly find and remediate sophisticated attacks that may have slipped through defenses leading to a significant reduction in time-to-detection (TTD) for Cisco customers compared to industry averages.

In this lab you will leverage the FMC to provision secure access for remote corporate users and configure secure connectivity between Main/HQ and Branch/Remote office sites. Using FMC, you will configure a Site to Site VPN between a Main Office and a Branch office location. Also, you will use FMC to provision Remote Corporate Users to connect to the Main Office and Branch location. The lab will showcase the various steps involved and new features available such as MFA and LDAP attributes.

Topology

This content includes preconfigured users and components to illustrate the scripted scenarios and features of the solution. Most components are fully configurable with predefined administrative user accounts. You can see the IP address and user account credentials to use to access a component by clicking the component icon in the **Topology** menu of your active dCloud session and in the scenario steps that require their use.

NOTE: For simplicity, not all IP addresses and VLANs are shown.

Figure 1. Lab Topology

Credentials

Login credentials are provided in line in the guide steps as needed. However, for your convenience, the following table lists the credentials used in these scenarios.

Table 2. Device Login Credentials

VM	Login	Password
FMC	admin	C1sco12345
All NGFWs	admin	C1sco12345
All Windows Workstations	Administrator	C1sco12345
Splunk	admin	C1sco12345
ISE and ISE PIC	admin	C1sco12345
Inside Linux Server	root	C1sco12345
Outside Linux Server	root	C1sco12345
CSR	admin	C1sco12345

There are also several users that have been created for purposes of passive authentication and RBAC.

Table 3. User Credentials

Login	Password	Details
dilbert	C1sco12345	Active Directory user in Engineering group
harry	C1sco12345	Active Directory user in HR group
ira	C1sco12345	Active Directory user in Finance and Investment groups
rita	C1sco12345	Active Directory user in IT group
alicia	C1sco12345	ISE local user, FDM Admin user
oliver	C1sco12345	ISE local user, FDM Read Only user
victoria	C1sco12345	ISE local user, VPN user
william	C1sco12345	ISE local user, FDM Read Write user

Get Started

BEFORE PRESENTING

Cisco dCloud strongly recommends that you perform the tasks in this document with an active session before presenting in front of a live audience. This will allow you to become familiar with the structure of the document and content.

It may be necessary to schedule a new session after following this guide in order to reset the environment to its original configuration.

PREPARATION IS KEY TO A SUCCESSFUL PRESENTATION.

Follow the steps to schedule a session of the content and configure your presentation environment.

1. Initiate your dCloud session. [\[Show Me How\]](#)

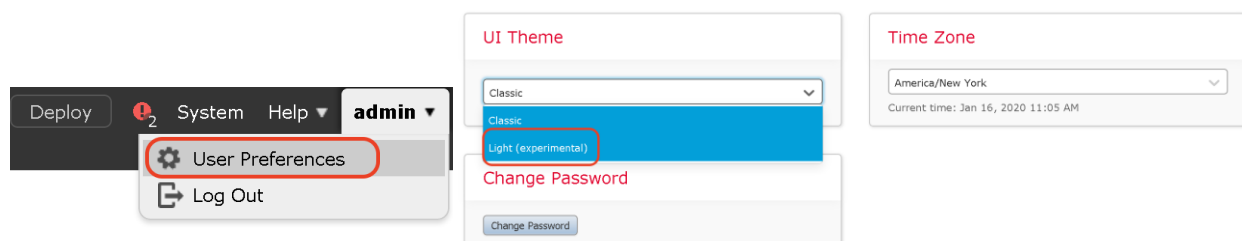
NOTE: It may take up to 10 minutes for your session to become active.

2. It is assumed in this guide that you are connecting to all devices from the Jumpbox RDP session, using the provided details.

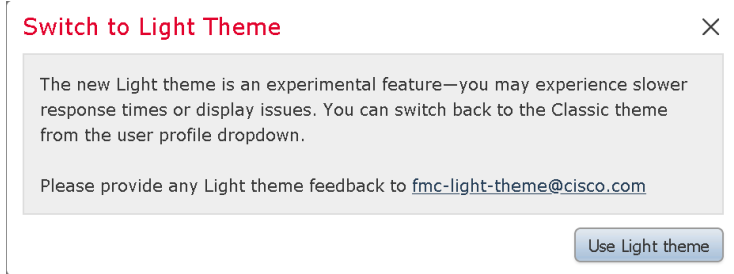
NOTE: You can also connect to the workstation with **Cisco AnyConnect VPN** [\[Show Me How\]](#) and the **local RDP client on your laptop** [\[Show Me How\]](#)

Jumpbox: **198.18.133.50**, Username: **administrator**, Password: **C1sco12345**

3. From the jumpbox, open **Firefox**. It should open with FMC login page as the default page. The login name and password will be prepopulated. Click **Login**
4. Navigate to **admin > User Preferences** in the top right-hand corner. Under the **General** tab, click on the dropdown under **UI Theme** and select **Light** theme.



5. A popup will show up with a disclaimer about the Light Theme. Click on **Use Light theme**.



6. The FMC UI will change from the classic UI to new UI.

The above step is optional for Firepower release 6.5. In Firepower release 6.6, the above UI theme is used by default. However, all the lab screenshots were captured using the new UI theme. It has no impact on the configuration steps.

NOTE: You can also connect to the workstation using the Cisco dCloud Remote Desktop client [\[Show Me How\]](#). The dCloud Remote Desktop client works best for accessing an active session with minimal interaction. However, many users experience connection and performance issues with this method.

7. For opening FDM, in the Jumpbox, open up a Firefox tab and click on **NGFW2 (FDM)** to proceed with this exercise. Login to FDM using the credentials **admin** and **C1sco12345**.

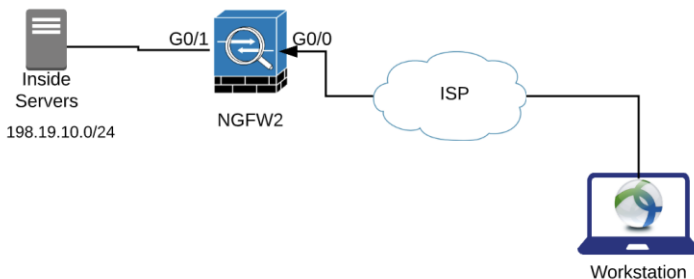
Scenario 1. Securing Remote Users for Corporate Access

Previously the Firepower Device Manager (FDM) provided a very basic RA VPN functionality. With 6.4, this functionality has been greatly expanded and is almost on par with the functionality present on the FMC. With the support of Client certificate authentication and double authentication (secondary authentication server) being added, we can configure RAVPN with multi-factor authentication and posture assessment (CoA using ISE). With 6.5, support for using DUO LDAP for Secondary Authentication was added. In this lab, we will be showcasing both the methods for integration with DUO MFA.

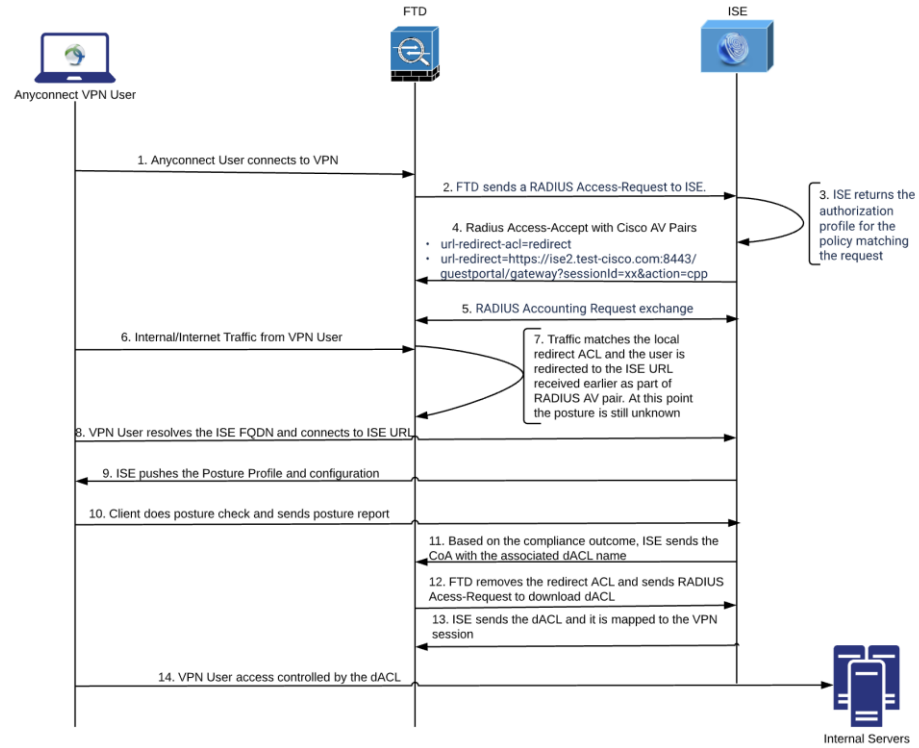
NOTE: You will be using ISE in this scenario. To save time – and avoid this turning into an ISE lab – ISE has been preconfigured. ISE-PIC is also included in this lab but is not completely configured.

The objective of this are:

- Create the objects required for the configuration of RA VPN with RADIUS Authentication and ISE posture assessment with Change of Authorization.
- Run the RA VPN setup wizard.
- Test RA VPN with ISE for RADIUS authentication and POSTURE assessment.



In this exercise, we will use ISE to validate the posture of the user machine. The flow of events will be as below :



Pre-Configuration Steps

Verify Licensing on the NGFW

1. After successful login to the FDM, it will land on the Device page. Navigate to **Smart License > View Configuration**.
2. You will see all the licenses already enabled. In an actual deployment, you would have to register to the Smart License Server and enable Export Control License.

Firepower Device Manager


Monitoring
Policies
Objects
Device: [ngfw2.dcloud.io...](#)**Interfaces**

Connected
Enabled 3 of 4

[View All Interfaces](#)**Routing**

3 routes

[View Configuration](#)**Updates**

Geolocation, Rule, VDB, System Upgrade,
Security Intelligence Feeds

[View Configuration](#)**Smart License**

Registered

[View Configuration](#)**Backup and Restore**[View Configuration](#)**Troubleshoot**

No files created yet

[REQUEST FILE TO BE CREATED](#)**Site-to-Site VPN**

There are no connections yet

[View Configuration](#)**Remote Access VPN**

Configured
2 connections | 3 Group Policies

[View Configuration](#)**Advanced Configuration**

Includes: FlexConfig, Smart CLI

[View Configuration](#)

Monitoring Policies Objects **Device: ngfw2.dcloud1...** admin Administrator

Threat DISABLE Enabled

This License allows you to perform intrusion detection and prevention and file control. You must have this license to apply intrusion policies in access rules. You also must have this license to apply file policies that control files based on file type.

Includes: Intrusion Policy

Malware DISABLE Enabled

This License allows you to perform Cisco Advanced Malware Protection (AMP) with AMP for Firepower and AMP Threat Grid. You must have this license to apply file policies that detect and block malware in files transmitted over your network.

Includes: File Policy

URL License DISABLE Enabled

This license allows you to control web access based on URL categories and reputations, rather than by individual URL alone. You must have this license to deploy access rules that filter web traffic based on category and reputation.

Includes: URL Reputation

RA VPN License Type PLUS DISABLE Enabled

Please select the license type that you purchased to enable remote access VPN. Note that Firepower Device Manager does not support any of the advanced features covered by the Apex license.

Includes: RA-VPN

PERPETUAL LICENSES INCLUDED

Base License ENABLED ALWAYS Enabled

This perpetual license is included with the purchase of the system. You must have this license to configure and use the device. It covers all features not covered by subscription licenses.

Verify objects and configuration needed for this scenario

NOTE: Most of these objects can be created while running the RA VPN wizard. This approach may be better for administrators that are not familiar with the components of the RA VPN configuration. However, in this scenario you will create the objects separately. This will simplify running the RA VPN wizard later.

1. Navigate to **Objects > Networks**. Verify the following objects are configured. You can enter the name of the object in the **Search** column.

Name	Type	Network
Corporate_LAN	Network	198.19.10.0/24
DNS_Server	Host	198.19.10.100
VPN_Pool	Network	198.19.10.64/29
ISE_Server	Host	198.19.10.130
DCloud_LAN1	Network	10.0.0.0/8

DCloud_LAN2	Network	198.19.255.0/24
DCloud_Jumpbox	Host	198.18.133.50

Network Objects and Groups

15 objects

Search

#	NAME	TYPE	VALUE	ACTIONS
1	Corporate_LAN	NETWORK	198.19.10.0/24	
2	DNS_Server	HOST	198.19.10.100	
3	GW123_1	HOST	198.18.133.111	
4	GW123_2	HOST	198.18.133.200	
5	Net123	NETWORK	1.2.3.0/24	
6	OutsideIPv4DefaultRoute	NETWORK	0.0.0.0/0	
7	OutsideIPv4Gateway	HOST	198.18.128.1	
8	VPN_Pool	NETWORK	198.19.10.64/29	
9	Webservers	NETWORK	198.19.10.200/30	
10	any-ipv4	NETWORK	0.0.0.0/0	
11	any-ipv6	NETWORK	::/0	
12	ISE_Server	HOST	198.19.10.130	
13	Dcloud_LAN1	NETWORK	10.0.0.0/8	
14	Dcloud_LAN2	NETWORK	198.19.255.0/24	
15	Dcloud_Jumpbox	HOST	198.18.133.50	

Note: Since there are multiple lab guides with different scenarios, it is possible that in some cases the highlighted objects might not be created. Please create the missing objects in this section.

2. Navigate to Ports and verify that the below port objects are created.
 - a. Search for **DNS**. An object **DNS over UDP** should be already configured. Verify the below details:
 - i. Name: DNS over UDP.
 - ii. Protocol: UDP
 - iii. Port: 53.

Port Objects and Groups

2 of 189 objects found

Search DNS

#	NAME	TYPE	VALUE	ACTIONS
1	DNS over TCP	TCP	53	
2	DNS over UDP	UDP	53	

- b. Search for **8443**. An object **HTTP_PORTS-8443** should be already configured. Verify the below details:
 - i. Name: HTTP_PORTS-8443
 - ii. Protocol: TCP

iii. Port: 8443

Port Objects and Groups

1 of 189 objects found

8443

#	NAME	TYPE	VALUE	ACTIONS
1	HTTP_PORTS-8443	TCP	8443	

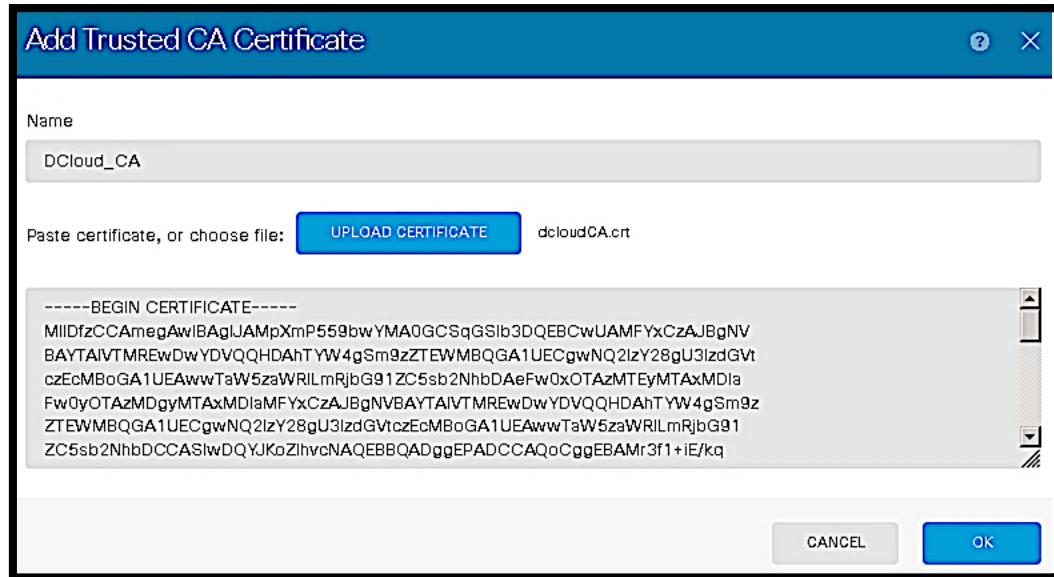
NOTE: We have added the above port objects to be eventually used in the ISE Posture redirect access-list.

- The DNS server has already been configured on NGFW2 for resolving ISE and DUO API URLs. Using **Putty** on the **Jumpbox**, login to the NGFW2 CLI using **admin/C1sco12345**. Run the command **show running-config dns**. The output should be as shown:

```
> show running-config dns
dns domain-lookup any
DNS server-group CiscoUmbrellaDNSServerGroup
  name-server 208.67.222.222
  name-server 208.67.220.220
DNS server-group CustomDNSServerGroup
  name-server 198.19.10.100
DNS server-group DefaultDNS
dns-group CustomDNSServerGroup
```

Upload Certificate for Anyconnect VPN connection

- Navigate to **Certificates**. Click on **+** and select **Add Trusted CA Certificate**.
 - Enter Name as **DCloud_CA**.
 - Click on **Upload Certificates**. Navigate to **Desktop > Certificates > Lab Certificates > Other Certificates** and select **dcloudCA.pem**
 - Click on **OK**.



Add Trusted CA Certificate

Name: DCloud_CA

Paste certificate, or choose file: UPLOAD CERTIFICATE dcloudCA.crt

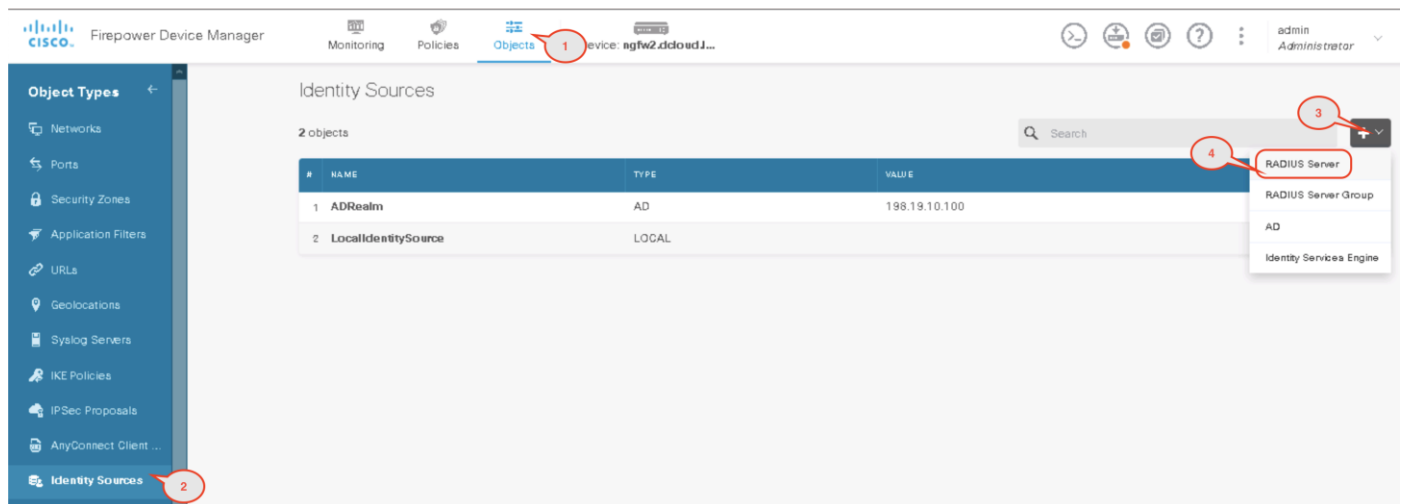
```
-----BEGIN CERTIFICATE-----
MIIDfzCCAmegAwIBAgIJAMpXmP559bwYMA0GCSqGSIb3DQEBCwUAMFYxCzAJBgNV
BAYTAiVTMREwDwYDVQQHDAhTYW4gSm9zZTEWMBQGA1UECgwNQ2l2Y28gU3lzdGVt
czEcMBoGA1UEAwwTaW5zaWRILmRjbG91ZC5sb2NhbDAeFw0xOTAzMTEyMTAxMDIa
Fw0yOTAzMDEyMTAxMDIaMFYxCzAJBgNVBAYTAiVTMREwDwYDVQQHDAhTYW4gSm9z
ZTEWMBQGA1UECgwNQ2l2Y28gU3lzdGVtczEcMBoGA1UEAwwTaW5zaWRILmRjbG91
ZC5sb2NhbDCCASlwdQYJKoZIhvcNAQEBBQADggEPADCCAQoCggEBAMr3f1+IE/kq
-----END CERTIFICATE-----
```

CANCEL OK

Create and configure a RADIUS server group for ISE

A RADIUS Server Group object can contain multiple RADIUS Servers. Here, we will create a RADIUS Server object and then create a RADIUS Server Group object with the RADIUS Server object as the RADIUS Server. It will be used for Authentication and Authorization of Remote Access VPN Users.

1. Navigate to **Objects > Identity Sources**. Click on **+** and select **RADIUS Server**



Firepower Device Manager

Monitoring Policies **Objects** 1 device: ngfw2.dcloud1...

Object Types

- Networks
- Ports
- Security Zones
- Application Filters
- URLs
- Geolocations
- Syslog Servers
- IKE Policies
- IPSec Proposals
- AnyConnect Client ...
- Identity Sources** 2

Identity Sources

2 objects

#	NAME	TYPE	VALUE
1	ADRealm	AD	198.19.10.100
2	LocalIdentitySource	LOCAL	

Search

3

4

- RADIUS Server
- RADIUS Server Group
- AD
- Identity Services Engine

- a. Enter Name as **ISE_Server**.
- b. IP Address as **198.19.10.130**.
- c. Server Secret Key as **C1sco12345**.
- d. Expand the **RA VPN Only** section. Click **Create new Extended Access List**.

Add RADIUS Server

Name
ISE_Server

Server Name or IP Address
198.19.10.130

Authentication Port
1812

Timeout ⓘ
10 seconds
1-300

Server Secret Key
••••••••••

RA VPN Only (if this object is used in RA VPN Configuration)

Redirect ACL

Filter

- WebserverAccess
- redirect

Create new Extended Access List

CANCEL OK

Create an extended access list called **redirect** that looks like the below screenshot. This will be used to determine what traffic is to be redirected to ISE while posture assessment is taking place. The ACEs where the action is block will be exempted from redirection. Some of the traffic such as DNS and ISE communication has to be exempted from redirection since we want the Anyconnect Client to be able to resolve the ISE redirect URL and communicate with ISE for posture config download (explained in the flow diagram).

Add Smart CLI Object

Name

redirect

Description

CLI Template

Extended Access List

Template

Access List Name

Show disabled
Reset

```

1 access-list redirect extended
2 configure access-list-entry deny
3 deny network source [any-ipv4] destination [ISE_Server]
4 configure deny port any-source
5 deny port source ANY destination [HTTP_PORTS-8443]
6 configure logging default
7 default log set log-level INFORMATIONAL log-interval 300
8 configure access-list-entry deny
9 deny network source [any-ipv4] destination [DNS_Server]
10 configure deny port any-source
11 deny port source ANY destination [DNS_UDP]
12 configure logging default
13 default log set log-level INFORMATIONAL log-interval 300
14 configure access-list-entry permit
15 permit network source [any-ipv4] destination [any-ipv4]
16 configure permit port any
17 permit port source ANY destination ANY
18 configure logging default
19 default log set log-level INFORMATIONAL log-interval 300

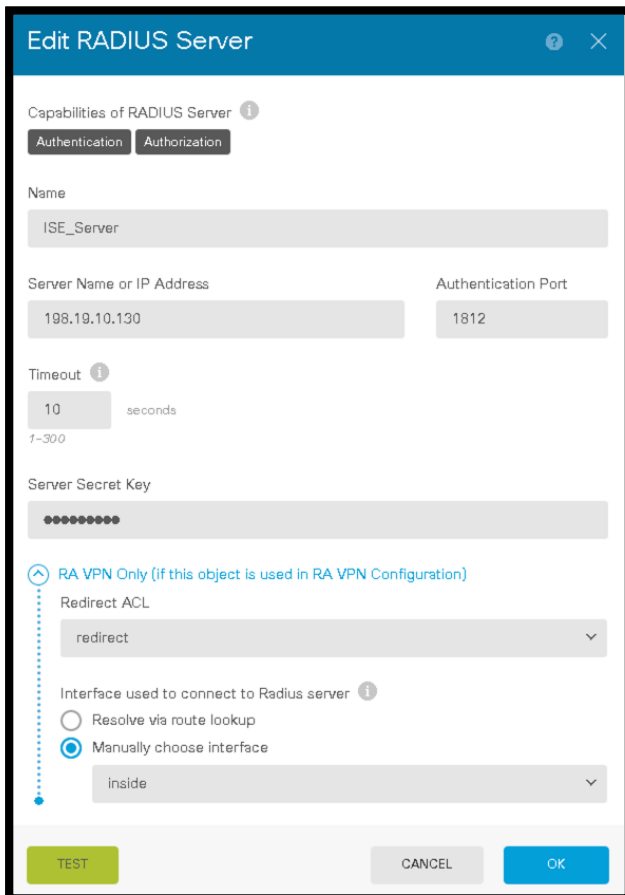
```

CANCEL

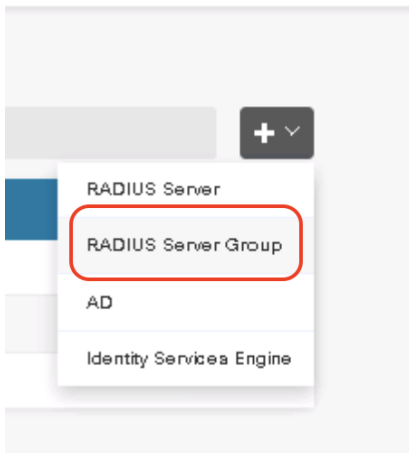
OK

- e. Set Name as **redirect**.
- f. In the template, enter the following configuration.
 - i. On Line 2, change action to deny. Lines 3-5 will appear.
 - ii. On Line 3, change source-network to any-ipv4 and destination-network to ISE_Server.
 - iii. On Line 4, change options to any-source. Line 5 will appear.
 - iv. On Line 5, change destination-port to HTTP_PORTS-8443.
 - v. On Line 6, change log-state to default. Line 7 will appear
 - vi. Go back to Line 2, click on the ... sign left to the configure access-list-entry command and click on Duplicate.
 - vii. On Line 8, change action to deny. Lines 9-11 will appear.
 - viii. On Line 9, change source-network to any-ipv4 and destination-network to DNS_Server.
 - ix. On Line 10, change options to any-source. Line 11 will appear.

- x. On Line 11, change destination-port to DNS_UDP.
 - xi. On Line 12, change log-state to default. Line 13 will appear.
 - xii. Go back to Line 8, click on the ... sign next to the configure access-list-entry command and click on Duplicate.
 - xiii. On Line 14, change action to permit. Lines 15-17 will appear.
 - xiv. On Line 15, change source-network to any-ipv4 and destination-network to any-ipv4.
 - xv. On Line 16, change options to any. Line 17 will appear.
 - xvi. On Line 18, change log-state to default. Line 19 will appear.
- g. Click on OK.
 - h. Set the Interface used to connect to Radius server as **Manually choose interface** and select in10 from the dropdown.
 - i. Click on **OK**



2. On the same object management page under **Identity Sources**, click on + and select **RADIUS Server Group**.

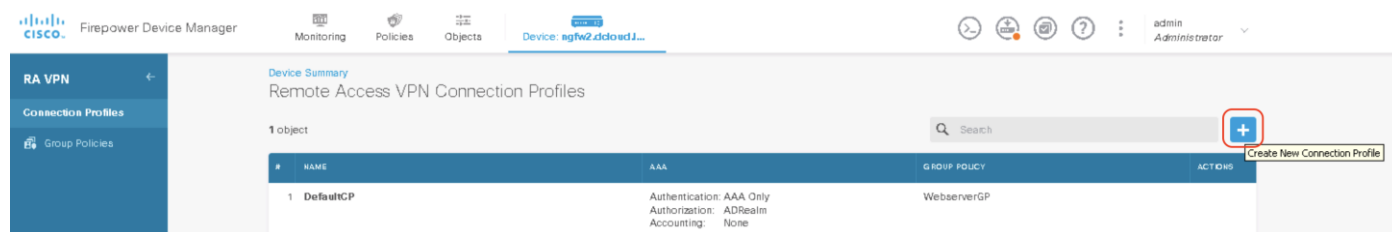


- a. Set the Name as **ISE_Server_Group**.
- b. Check Dynamic Authorization.
- c. Click on **+** under **RADIUS Server** and select the **ISE_Server** object we just created.
- d. Click on **OK**.

A screenshot of a "Add RADIUS Server Group" dialog box. The dialog has a blue header with the title and a close button. The main area contains several fields: "Name" with the value "ISE_Server_Group"; "Dead Time" set to "10" minutes; "Maximum Failed Attempts" set to "3"; a checked checkbox for "Dynamic Authorization (for RA VPN only)"; "Port" set to "1700"; and a "Realm that Supports the RADIUS Server" dropdown menu currently showing "Please select". Below these is a section for "RADIUS Server" with a blue informational message: "The servers in the group should be backups of each other". Under this message is a plus icon and a list of servers, with "1. ISE_Server" selected and a "Test" link next to it. At the bottom of the dialog are three buttons: "TEST ALL SERVERS..." (green), "CANCEL" (gray), and "OK" (blue).

Run Remote Access VPN Wizard

In the FDM, navigate to **Device > Remote Access VPN > View Configuration**. Click **+** icon.



Complete the Connection Profile page of the wizard.

1. Set Connection Profile Name as **RAVPN**. The Group Alias will automatically get updated.

Connection and Client Configuration

Specify how to authenticate remote users and the AnyConnect clients they can use to connect to the inside network.

Connection Profile Name

This name is configured as a connection alias, it can be used to connect to the VPN gateway

RAVPN

Group Alias

RAVPN

[Add Group Alias](#)

Group URL

[Add Group URL](#)

Choosing the authentication method for VPN Client

In the next page, the RA VPN wizard offers various methods of authentication - Primary Authentication using:

AAA only – Username/password authentication using RADIUS, LDAP or AD. In this lab ISE is used as RADIUS server to authenticate the VPN user credentials

Client Certificate Only – Each user is authenticated with a client certificate. The client certificate must be configured on VPN client endpoints. By default, the username is derived from the client certificate fields CN and OU. In this lab, the Anyconnect client machine already has device certificate installed and they are used to authenticate the identity of the VPN user. By default, the CN from the device certificate will be used to populate the username field of the authentication prompt in the Anyconnect Client

AAA and Client Certificate – A combination of ISE as RADIUS server and client certificate.

Note: The RA VPN configuration also offers Secondary authentication method. For example, you can set the primary authentication server to an LDAP or Active Directory realm and the secondary authentication to a RADIUS server. In this lab we will be using the Secondary Authentication method for MFA using DUO.

2. For **Primary Authentication Source** authentication type, you can select **AAA Only**, **Client Certificate Only** or **AAA and Client Certificate**. However, for this lab exercise, we will use the method **AAA and Client Certificate**.
 - a. Under **Primary Authentication Source for User Authentication** select **ISE_Server_Group** (Authentication Server is available only if the **Authentication Type** is AAA only or AAA and Client Certificate).
 - b. Let the **Username from Certificate** field be as is to allow for pre-filling the username from the certificate's CN. Expand **Advanced** and select **Prefill username from certificate on user login window**.

Primary Identity Source

Authentication Type

☐ AAA Only
 ☐ Client Certificate Only
 ☒ AAA and Client Certificate

Primary Identity Source for User Authentication: ISE_Server_Group

Fallback Local Identity Source: Please Select Local Identity Source

☐ Strip Identity Source server from username
☐ Strip Group from Username

Username from Certificate

☒ Map Specific Field
 ☐ Use entire DN (distinguished name) as username

Primary Field: CN (Common Name)
 Secondary Field: OU (Organisational Unit)

Advanced

☒ Prefill username from certificate on user login window
☐ Hide username in login window

3. Set Authorization and Accounting Server to **ISE_Server_Group**.
 - a. Click on **+** under **IPv4 Address Pool** and select **VPN_Pool**. Click **Ok**.
 - b. Click on **Next**.

Authorization Server: Please select

Accounting Server: ISE_Server_Group

Client Address Pool Assignment

IPv4 Address Pool
Endpoints are provided an address from this pool

+
VPN_Pool

IPv6 Address Pool
Endpoints are provided an address from this pool

+

DHCP Servers

+

Complete the Remote User Experience Page

- Under **View Group Policy**, select **DfltGrpPolicy** from the drop-down menu. Click on **Edit**

Monitoring Policies Objects Device: agfw2.dcloud1...

Remote Access VPN

1 Connection and Client Configuration 2 Remote User Experience 3 Global Settings 4 Summary

Remote Users AnyConnect Clients Internet Client Certificate OUTSIDE GigabitEthernet INSIDE INTERFACES Corporate Resources

Identity Source for User Authentication

Remote User Experience

A group policy is a collection of user-oriented session attributes which are assigned to client when a VPN connection is established. Select or create a Group Policy object.

View Group Policy

DfltGrpPolicy

Policy Group Brief Details

DNS Server CustomDNSServerGroup

BACK NEXT

- Under **DNS Server**, select **CustomDNSServerGroup** under DNS Server.
- Under **Corporate Resources (Split Tunneling)**, for **IPv4 Split Tunneling** select **Exclude networks specified below**.
 - Click on **+** under **IPv4 Split Tunneling Networks** and select **DCloud-LAN1**, **DCloud-LAN2** and **DCloud-Jumpbox**. You can select more than one options by pressing **Control** on the keyboard.
 - Click **OK**.

Corporate Resources (Split Tunneling)

IPv4 Split Tunneling

Exclude networks specified below

IPv6 Split Tunneling

Allow all traffic over tunnel

IPv4 Split Tunneling Networks



Filter

- Corporate_LAN Network
- ☒ DCloud_Jumpbox Host
- ☒ DCloud_LAN1 Network
- ☒ DCloud_LAN2 Network
- DNS_Server Host

Create new Network CANCEL OK

4. Under **AnyConnect Client**, click on + and click **Create new AnyConnect Profile**.

AnyConnect Client

AnyConnect Client Profiles *(optional)*
If you configure a fully-qualified domain name for the outside interface, a default profile will be created for you. Alternatively, you can upload your own client profile.

+ Filter

- ACProfileAllowRemote
- AnyConnectProfile

Create new AnyConnect Client Profile CANCEL OK

Add AnyConnect Client Profile

Name
AnyConnectProfile

Description

AnyConnect Client Profile

UPLOAD AnyConnectProfile.xml

CANCEL OK

- a. Click on **Upload**. Navigate to **Desktop > RA VPN** and select **AnyConnectProfile.xml**. Click **OK**
- b. Select the newly created profile named **AnyconnectProfile**. Click on **OK**.

5. Click **Next**

Complete Global Settings Page

1. Under **Certificate of Device Identity**, click **Create New Internal Certificate**.

- a. Click on **Upload Certificate and Key**.
- b. Enter Name as **NGFW2_Outside**.
- c. Click on **Upload Certificate**. Navigate to **Desktop > Certificates > Lab Certificates** and select **ngfw-dcloud.crt**
- d. Click on **Upload Key**. Navigate to **Desktop > Certificates > Lab Certificates** and select **ngfw-dcloud.key**
- e. Click on **OK**.
- f. Select the newly created Internal Certificate Object **NGFW2_Outside**.

Add Internal Certificate

Name
NGFW2_Outside

SERVER CERTIFICATE (USER AGENT)

Paste certificate, or choose file: **UPLOAD CERTIFICATE** ngfw-dcloud.crt

```
-----BEGIN CERTIFICATE-----
MIIDZzCCAreGAWlBAGlJAKdbAbuuBzPOMA0GCSqGSIb3DQEBCwUAMFYxCzAJBgNV
BAYTAiVTMREwDwYDVQQHDAhTYW4gSm9zZTEwMBQGA1UECgwNQ2lyZ28gU3lzdGVt
czEcMBoGA1UEAwwTaW5zaWRILmRjbG91ZC5sb2NhbDAeFw0xOTAzMTEyMTU5MDZa
Fw0yNDZzMDEyMTU5MDZaMFQxCzAJBgNVBAYTAiVTMREwDwYDVQQHDAhTYW4gSm9z
-----END CERTIFICATE-----
```

CERTIFICATE KEY

Paste key, or choose file: **UPLOAD KEY** ngfw-dcloud.key

```
-----BEGIN RSA PRIVATE KEY-----
MIIEpQIBAAKCAQEA0t3qL6tBMW7axEKvJ3pKtq1O2alQ925IAN0DcJCskGCoCSTG
BsmDfQd8yB60em5/D5ymTh0C8Lu9HjYkOVYnnwC3QSRwlQWRnhNRIQpcUSBr0u3
7v3uTFyRNIMgQtDnyzdXxAC7D2mw4EsxhGGL5fWVWVWO4pPVfJHvWxZpTBSLra
MVQNNIt4u5xMyQUTck7G87AZyUltqBhYFH90TEgwcYkYFIMTg7xWCNDc+k4p4Qg4
-----END RSA PRIVATE KEY-----
```

CANCEL OK

2. Set Outside Interface to **outside**.
3. Enter **ngfw2-outside.dcloud.local** as Fully-qualified Domain Name for the Outside Interface.

Monitoring Policies Objects Device: **ngfw2.dcloud.local**

Remote Access VPN

1 Connection and Client Configuration 2 Remote User Experience 3 Global Settings 4 Summary

Global Settings

These settings control the basic functioning of the connection. Changes to any of these options apply to all connection profiles; you cannot configure different settings in different profiles.

Certificate of Device Identity: **NGFW2_Outside1**

Outside Interface: **outside (GigabitEthernet0/0)**

Fully-qualified Domain Name for the Outside Interface: **ngfw2-outside.dcloud.local**
e.g. ns.vpn.example.com

Access Control for VPN Traffic: Decrypted VPN traffic is subjected to access control policy inspection by default. Enabling the Bypass Access Control policy for decrypted traffic option bypasses the access control policy, but for remote

Bypass Access Control policy BACK NEXT

4. Configure NAT exemption

- Click on **+** under NAT Exempt Inside Interfaces and select **in10**.
- Click on **+** under NAT Exempt Inside Networks and select **Corporate_LAN**.

5. Configure AnyConnect Image to be pushed.

- Under AnyConnect Package, click on **Upload Package**.
- Select **Windows**. Navigate to **Desktop > RA VPN** and select **anyconnect-win-4.7.01076-webdeploy-k9.pkg**.
- Click on **Open**.
- Once the AnyConnect Package gets uploaded, click on **Next**

Monitoring
Policies
Objects

Device: sgfw2.dcloud.j...

☐ Bypass Access Control policy for decrypted traffic (sysopt permit-vpn)

NAT Exempt

☒

Inside Interfaces

The interfaces through which remote access VPN users can connect to the internal networks

+

Inside Networks

The internal networks remote access VPN users are allowed to use. The IP versions of the internal networks and address pools must match, either IPv4, IPv6, or both.

+

AnyConnect Package

If a user does not already have the right AnyConnect package installed, the system will launch the AnyConnect installer when the client authenticates for the first time. The user can then install the package from the system.

You can download AnyConnect packages from software.cisco.com. You must have the necessary AnyConnect software license.

Packages

UPLOAD PACKAGE

BACK
NEXT

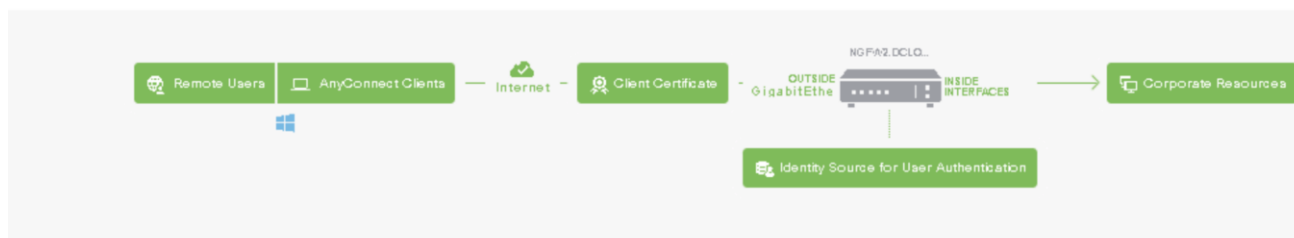
Note: We are not selecting the option 'Bypass Access Control Policy for decrypted traffic'. If you select this option, we don't need to create specific rules in the Access Control Policy to allow the decrypted VPN traffic. For the purpose of this lab, we will be modifying the Access Control to allow the VPN traffic after decryption.

Review the Configuration on the **Summary** page and click **Finish**.

Click on **Finish**.

Remote Access VPN

- 1 Connection and Client Configuration 2 Remote User Experience 3 Global Settings 4 **Summary**



^ Summary

Review the summary of the Remote Access VPN configuration.

RAVPN

STEP 1: CONNECTION AND CLIENT CONFIGURATION

Primary Identity Source

Authentication Type AAA and Client Certificate

Primary Identity Source ISE_Server_Group

Fallback Local Identity Source -

Strip Identity Source server from username No

Strip Group from Username No

Username from Certificate Map Specific Field

Primary Field CN (Common Name)

Secondary Field OU (Organisational Unit)

Advanced

Secondary Identity Source

Secondary Identity Source for User Authentication -

Fallback Local Identity Source -

Advanced

Authorization Server ISE_Server_Group

Accounting Server ISE_Server_Group

Client Address Pool Assignment

IPv4 Address Pool VPN_Pool

IPv6 Address Pool -

DHCP Servers -

STEP 2: GROUP POLICY

Group Policy Name	DftGrpPolicy
Banner + DNS Server	
DNS Server	CustomDNSServerGroup
Banner text for authenticated clients	-
Session Settings	
Maximum Connection Time / Alert Interval	Unlimited / 1 Minutes
Idle Timeout / Alert Interval	30 / 1 Minutes
Simultaneous Login per User	3
Split Tunneling	
IPv4 Split Tunneling	Exclude networks specified below
IPv4 Split Tunneling Networks	DCloud_Jumpbox DCloud_LAN1 DCloud_LAN2
IPv6 Split Tunneling	Allow all traffic over tunnel
AnyConnect Client	
AnyConnect Client Profiles	AnyConnectProfile

STEP 3: GLOBAL SETTINGS

Certificate of Device Identity	NGFW2_Outside1
Outside Interface	GigabitEthernet0/0 (outside)
Fully-qualified Domain Name for the Outside Interface	ngfw2-outside.dcloud.local
Access Control for VPN Traffic	No
NAT Exempt	
NAT Exempt	Yes
Inside Interfaces	GigabitEthernet0/1 (inside)
Inside Networks	Corporate_LAN
AnyConnect Package	
Packages	Windows: anyconnect-win-4.7.02036-webdeploy-k9.pkg

Instructions

Instructions for your End Users to start connecting

Adding Secondary Authentication for Remote Access VPN Users using DUO MFA (Optional)

This section is optional. If you don't wish to add a Secondary authentication mechanism, please proceed with the steps in the section titled **Allowing VPN Traffic through the NGFW2**.

1. This section showcases two methods of DUO integrations. You can do either of the next two sections depending on the type of Duo Integration you want to implement.
 - a. Section A - Create and configure an LDAP server to connect to Duo. This method uses a direct LDAPS connection between the firewall and DUO Cloud. For the LDAPS method, you will need to keep the Integration Key, Secret Key and API Hostname from the Duo Application Dashboard handy.
 - b. Section B - Create and configure a RADIUS Server object for Duo Authentication Proxy. This method makes use of the Authentication Proxy server that was already set up for the purposes of the lab.
2. Complete the DUO account setup steps in Appendix A. This is done irrespective of the type of DUO integration being used.
3. Proceed to either of the next Section based on the choice of DUO integration.

Note: Skip this section and jump to Section B, if you wish to use DUO Authentication Proxy for RADIUS Authentication

Section A - Create and configure an LDAP server to connect to Duo

We will use the FTD API to create the Duo LDAP object. We can either use the API Explorer, POSTMAN, curl or write a custom client application, to create the object. The following procedure explains how to create the object using the FDM API Explorer.

Create LDAP Server object for DUO MFA

1. Click on more option button  and choose **API explorer**.
2. Click on the **DuoLDAPIdentitySource** resource to list the methods available.
3. Click on the **POST /object/duoldapidentitysources** method. The POST method is used to create a new LDAP identity source with the information present in the payload as JSON.
4. You can view the sample JSON payload under **Example Value**. Click on it and the same example values will be loaded onto the **body** under the Parameters. Now these values can be edited as per the requirements of this lab and then configured through the API call.
5. Modify the payload using the body value edit box as below:

```
{
  "name": "Duo-LDAP",
  "description": "Duo LDAP server for RA VPN",
  "apiHostname": "API-XXXXXXXXX.DUOSECURITY.COM",
  "port": 636,
  "timeout": 120,
  "integrationKey": "XXXXXXXXXXXXXXXXXXXX",
  "secretKey": "123456789",
  "type": "duoldapidentitysource"
}
```

The following parameters values were deleted **version**, **id** and **interface**. (These attributes are needed for PUT calls but not for POST.) The **apiHostname**, **integrationKey** and **secretKey** can be obtained from the DUO account page (Appendix A).

6. Click the **Try It Out!** button. The API explorer will show the equivalent curl command, the request URL the response body and the response code. The response code should show 200. If it is not 200, most likely you made an error, look at the response body for an error message. You can correct the body value and try again.
7. Return to FDM by clicking **Device** in the top menu.
8. Click **Objects**, then click **Identity Sources** in the table of contents. Your Duo LDAP object should appear in the list.

Note that you can delete the object using FDM, but you cannot edit it or see its contents. You must use the API for those actions.

The relevant methods are shown in the **DuoLDAPIdentitySource** group.

Upload the Trusted CA Certificate of DUO on FDM

1. On the Jumpbox, using Firefox navigate to the website <https://www.digicert.com/digicert-root-certificates.htm#roots>
2. Search for **DigiCert High Assurance EV Root CA** and download the certificate file by right-clicking on **Download PEM** and select **Save Link as**. Save the file in the **Downloads** folder.
3. In FDM, choose to **Objects > Certificates**.
4. Click **+ > Add Trusted CA Certificate**.
5. Enter the name for the certificate, **DigiCert_High_Assurance_EV_Root_CA**.
6. Click **Upload Certificate** and select the file you downloaded, **DigiCertHighAssuranceEVRootCA.crt.pem**

Add Trusted CA Certificate

Name

DigiCert_High_Assurance_EV_Root_CA

Paste certificate, or choose file:

UPLOAD CERTIFICATE

DigiCertHighAssuranceEVRootCA.cer

-----BEGIN CERTIFICATE-----

```
MIIDxTCCAq2gAwIBAgIQAgxcJmoLQJuPC3nyrkYldzANBgkqhkiG9w0BAQUFADBs
MQswCQYDVQQGEwJVUzEVMBMGA1UEChMMRGlnaUNlcnQgSW5jMRkwFwYDVQQLExB3
d3cuZGlnaWNlcnQuY29tMSswKQYDVQQDEyJEaWdpd3cuY29tY2VydCBlaWdoIEFz
c3VyYW5jZSBFViBSb290IENBMGM4XDTA2MTEwMDAwMDAwMFoXDTMxMTEwMDAwMDAw
MAkGA1UEBhMCVVMxFTATBgNVBAoTDERpZ2IDZXJ0IEluYzEZMBcGA1UECxmQd3d3
LWJzZ29tY2VydCBlaWdoIEFz
```

CANCEL

OK

7. Click **OK**.

Section B - Create and configure a RADIUS Server object for Duo Authentication Proxy

1. For convenience, we have already configured the Duo Authentication Proxy application on the internal Active Directory Server. You can open an RDP session to this server to take a look at the Auth Proxy configuration file. This file is located at **C:\Program Files (x86)\Duo Security Authentication Proxy\conf.**

```
; Complete documentation about the Duo Auth Proxy can be found
here:
; https://duo.com/docs/authproxy_reference

; MAIN: Include this section to specify global configuration
options.
; Reference: https://duo.com/docs/authproxy_reference#main-
section
; [main]

; CLIENTS: Include one or more of the following configuration
sections.
; To configure more than one client configuration of the same
type, append a
; number to the section name (e.g. [ad_client2])

[duo_only_client]

; SERVERS: Include one or more of the following configuration
sections.
; To configure more than one server configuration of the same
type, append a
; number to the section name (e.g. radius_server_auto1,
radius_server_auto2)

[radius_server_auto]
ikey=DIB86UGGW1KH44Y7LJCH
skey=aHGHby4M4f8crnmwDuXzQ80MOADtHxgtRJcmzq7lO
api_host=api-29acc416.duosecurity.com
radius_ip_1=198.19.10.1
radius_secret_1=C1sco12345
radius_ip_2=198.19.10.2
radius_secret_2=C1sco12345
radius_ip_3=198.19.10.3
radius_secret_3=C1sco12345
failmode=safe
client=duo_only_client
```

2. Update the configuration file with the values from the DUO Account noted earlier for the Application – **Firepower Threat Defense VPN**
 - a. **ikey** = Integration Key
 - b. **skey** = Secret Key
 - c. **api** = API hostname
 - d. Save and close the file once done.
3. On the AD Server, open up **Services.msc** and restart services for **Duo Authentication Proxy**. Make sure that this is successful.
4. Navigate to **C:\Program Files (x86)\Duo Security Authentication Proxy\bin** and run **authproxy_connectivity_tool**.
5. To see if the connection tests were successful, navigate to **C:\Program Files (x86)\Duo Security Authentication Proxy\log** and open the **connectivity_tool** file. Scroll to the bottom of the file and confirm that you see the following logs:

```

2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] Testing section 'duo_only_client' with
configuration:
2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] {}
2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] There are no configuration problems
2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] -----
2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] Testing section 'radius_server_auto' with
configuration:
2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] {'api_host': 'api-38238671.duosecurity.com',
'client': 'duo_only_client',
'failmode': 'safe',
'key': 'DI4RHEKLY5WQ0JO8EZ8U',
'port': '1812',
'radius_ip_1': '198.19.10.1',
'radius_ip_2': '198.19.10.2',
'radius_ip_3': '198.19.10.3',
'radius_secret_1': '*****',
'radius_secret_2': '*****',
'radius_secret_3': '*****',
'skey': '*****[40]'}
2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] There are no configuration problems
2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] -----
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] Testing section 'duo_only_client' with
configuration:
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] {}
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] No testing to be done for section.
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] -----
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] Testing section 'radius_server_auto' with
configuration:
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] {'api_host': 'api-38238671.duosecurity.com',
'client': 'duo only client',
'failmode': 'safe',
'key': 'DI4RHEKLY5WQ0JO8EZ8U',
'port': '1812',
'radius_ip_1': '198.19.10.1',
'radius_ip_2': '198.19.10.2',
'radius_ip_3': '198.19.10.3',
'radius_secret_1': '*****',
'radius_secret_2': '*****',
'radius_secret_3': '*****',
'skey': '*****[40]'}
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] The RADIUS Server has no connectivity problems.
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] -----
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] SUMMARY
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] No issues detected

```

6. Navigate to **Objects > Identity Sources**. Click on **+** and select **RADIUS Server**

- a. Enter Name as **Duo_Server**.
- b. IP Address as **198.19.10.100**.
- c. Server Secret Key as **C1sco12345**.
- d. Expand the **RA VPN Only** section. Set the Interface used to connect to Radius server as **Manually choose interface** and select in10 from the dropdown.

- e. Click on **OK**

The screenshot shows a configuration window titled "Add RADIUS Server". It contains the following fields and options:

- Name:** A text field containing "Duo_Server".
- Server Name or IP Address:** A text field containing "198.19.10.100".
- Authentication Port:** A text field containing "1812".
- Timeout:** A text field containing "10" with a unit of "seconds" and a range "1-300" below it.
- Server Secret Key:** A text field filled with dots.
- RA VPN Only (if this object is used in RA VPN Configuration):** A section header with a blue arrow icon.
- Redirect ACL:** A dropdown menu with "Please select" and a downward arrow.
- Interface used to connect to Radius server:** A section header with an information icon.
- Resolution Options:** Two radio buttons: "Resolve via route lookup" (unselected) and "Manually choose interface" (selected).
- Interface Selection:** A dropdown menu showing "inside" with a downward arrow.
- Buttons:** "CANCEL" and "OK" buttons at the bottom right.

7. Click on **+** and select **RADIUS Server Group**.
- Set the Name as **Duo_Server_Group**.
 - Click on **+** under **RADIUS Server** and select the **Duo_Server** object we just created.
 - Click on **OK**

NOTE: Test will fail since it uses the management interface.

Duo configuration for the RAVPN profile

We will be adding DUO MFA now as the secondary authentication server.

1. Navigate to **Device > Remote Access VPN > View Configurations**
2. Click **View** icon for the profile named **RAVPN**

#	NAME	AAA	GROUP POLICY	ACTIONS
1	DefaultCP	Authentication: AAA Only Authorization: ADRealm Accounting: None	WebserverGP	
2	RAVPN	Authentication: AAA and Client Certificate Authorization: ISE_Server_Group Accounting: ISE_Server_Group	DfltGrpPolicy	 

3. Click **Edit**. You should land at the Connection and Client Configuration page of the wizard.
4. Scroll down to **Secondary Identity Source**. Select **Duo_Server_Group** or **Duo_LDAP** (depending on the method of DUO integration chosen above). Expand **Advanced**.

- i. Since we will be pre-filling the username from the certificate's CN, let the Username from Certificate field be as is.
- ii. Select Prefill username from certificate on user login window. Select Hide username in login window.

Secondary Identity Source

Secondary Identity Source for User Authentication

Duo_Server_Group

Advanced

Fallback Local Identity Source for Secondary

Please Select Local Identity Source

☐ Use Primary username for Secondary login

Username from Certificate

☒ Map Specific Field

Primary Field *i* Secondary Field

CN (Common Name) OU (Organisational Unit)

☐ Use entire DN (distinguished name) as username

☒ Prefill username from certificate on user login window

☒ Hide username in login window

Username for session server

☒ Primary ☐ Secondary

Password Type

Prompt

Allowing VPN Traffic through the NGFW2

Modify the Access Control Policy to allow decrypted VPN traffic

1. Navigate to **Access Control** and click on +.
 - a. For the new rule, set Order to 1.
 - b. Name will be **RAVPN_Access** and Action will be **Allow**.
 - c. Set Source Zone to **outside_zone** and Source Networks to **VPN_Pool**.

Add Access Rule

Order	Title	Action
1	RAVPN_Access	Allow

Source/Destination Applications URLs Users ¹ **Intrusion Policy** File policy Logging

SOURCE

Zones ⁺ Networks ⁺ Ports ⁺

outside_zone VPN_Pool ANY

DESTINATION

Zones ⁺ Networks ⁺ Ports/Protocols ⁺

ANY ANY ANY

2. Navigate to **Intrusion Policy**. Enable Intrusion Policy and set the Level to **Balanced Security and Connectivity**.

Add Access Rule

Order	Title	Action
1	RAVPN_Access	Allow

Source/Destination Applications URLs Users ¹ **Intrusion Policy** File policy Logging

INTRUSION POLICY ☒

Level of Intrusion Policy
Balanced Security and Connectivity

Balanced Security and Connectivity
This policy is designed to balance overall network performance with network infrastructure security. This policy is appropriate for most networks. Select this policy for most situations where you want to apply intrusion prevention.

PREVENTING INTRUSIONS
Use intrusion policies as a last line of defense against unwanted traffic that you are otherwise allowing. An intrusion policy examines decoded packets for intrusions, exploits, and other attacks based on patterns, and can block or alter malicious traffic. Cisco delivers several intrusion policies with the Firepower system. These policies are designed by the Cisco Talos Security Intelligence and Research Group, who set the intrusion and preprocessor rule states and advanced settings.

3. Navigate to **File Policy**. Select **Block Malware All**.

Add Access Rule

Order	Title	Action
1	RAVPN_Access	Allow

Source/Destination Applications URLs Users ¹ Intrusion Policy **File policy** Logging

SELECT THE FILE POLICY
Block Malware All

CONTROLLING FILES AND MALWARE
Use file policies to detect malicious software, or malware, using Advanced Malware Protection for Firepower (AMP for Firepower.) You can also use file policies to perform file control, which allows control over all files of a specific type regardless of whether the files contain malware.

4. Navigate to **Logging**. Select **at Beginning and End of Connection** for Select Log Action. Click on **OK**.

Add Access Rule

Order	Title	Action
1	RAVPN_Access	Allow

Source/Destination Applications URLs Users **Logging** Intrusion Policy File policy

SELECT LOG ACTION

☒ At Beginning and End of Connection
☐ At End of Connection
☐ No Connection Logging

SEND CONNECTION EVENTS TO:

Select a Syslog Alert Configuration...

FILE EVENTS

☒ Log Files

Show Diagram ☐

CANCEL OK

NOTE: The logging configuration is for lab purposes. In most scenarios, the recommended logging configuration is to enable logging 'At End of Connection' to avoid generating the connection event twice.

Modify the NAT configuration to allow Posture traffic from Anyconnect Client

1. Navigate to **Policies > NAT**. To allow for Posture Assessment to trigger, we will modify the default **InsideOutsideNatRule** to allow VPN traffic also.
2. Edit the **InsideOutsideNatRule**. Change the **Source Interface** from **Inside** to **Any**. Click on **OK**.

Edit NAT Rule

?

×

Title

InsideOutsideNatRule

Create Rule for

Manual NAT

Status

☒

Manual NAT rules allow the translation of the source as well as the destination address of a network packet. Destination and port translation are optional. You can place manual NAT rules either before or after Auto NAT rules and insert the rules at a specific location.

Placement

Before Auto NAT Rules

Type

Dynamic

Packet Translation

Advanced Options

ORIGINAL PACKET

Source Interface

Any

Source Address

any-ipv4

Source Port

Any

Destination Address

Any

Destination Port

Any

TRANSLATED PACKET

Destination Interface

outside

Source Address

Interface

Source Port

Any

Destination Address

Any

Destination Port

Any

Show Diagram

☐

CANCEL

OK

NOTE: The FDM adds in a NAT exemption statement for VPN traffic which is only visible via CLI. However, this last change was done to allow for Posture Assessment to trigger. As Posture probes from Anyconnect Client will be internet bound to enroll.cisco.com, the traffic has to hit the NGFW outside interface and do a u-turn. This modification will ensure that the probes are translated to the outside interface IP for internet routing.

Deploy and TEST VPN

Deploy and verify the NGFW configuration

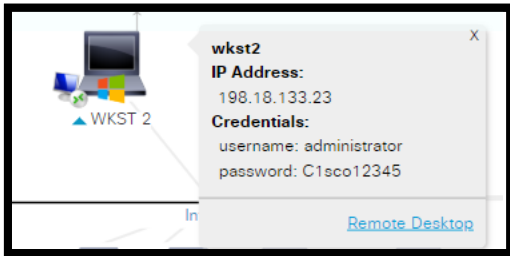
NOTE: In this scenario, the definition of a compliant system is a system that has a file called compliant.txt on the desktop. If you have done the previous scenario, you will directly be found compliant, so you can ignore Steps 6-9. If this is your first RA VPN scenario, then Wkst2 will start out as non-compliant. And furthermore, the posture module is already installed on Wkst2.

1. Deploy the configuration on **NGFW2** and wait until the deployment completes
2. You can verify the configuration from the NGFW CLI

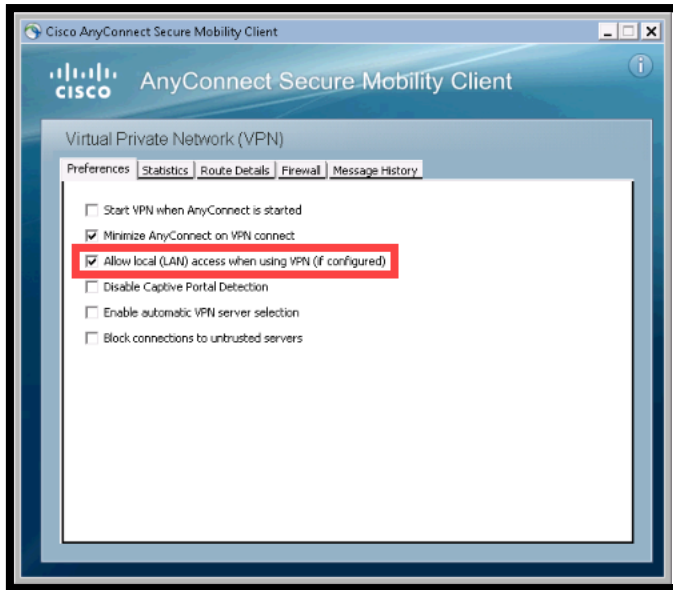
3. After the deployment is complete, from the Jumpbox open a PUTTY session to the **NGFW2 CLI**. Log in as **admin/C1sco12345** and run some or all of the following commands:
 - a. `show running-config tunnel-group`
 - b. `show running-config group-policy`
 - c. `show running-config crypto`
 - d. `show running-config ip local pool`
 - e. `show run nat`
4. Test AAA by running the following command on the NGFW2 CLI:
test aaa-server authentication ISE_RADIUS host 198.19.10.130 username vpnuser password C1sco12345

Connect with Anyconnect

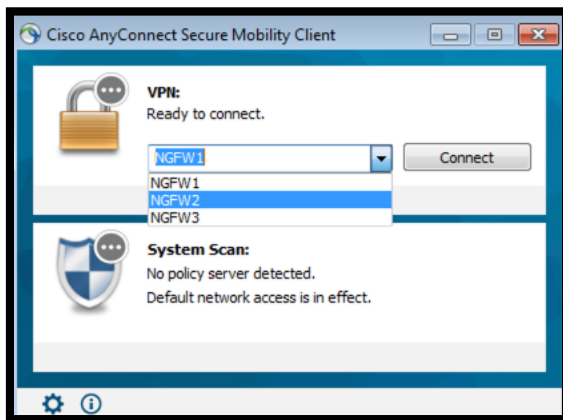
1. Connect to **Wkst2**. You will automatically be logged in as the administrator. You can connect using one of two methods.
 - a. From the topology map, as show in this picture. This is the recommended method.



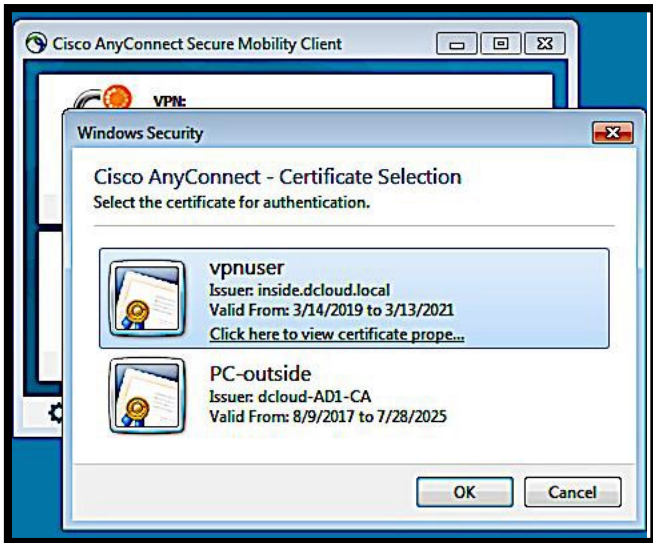
- b. By clicking the Wkst2 (Outside PC) shortcut in the Remote Desktops folder the Jumpbox desktop. However. If you do this, you must allow local LAN access in the AnyConnect client.



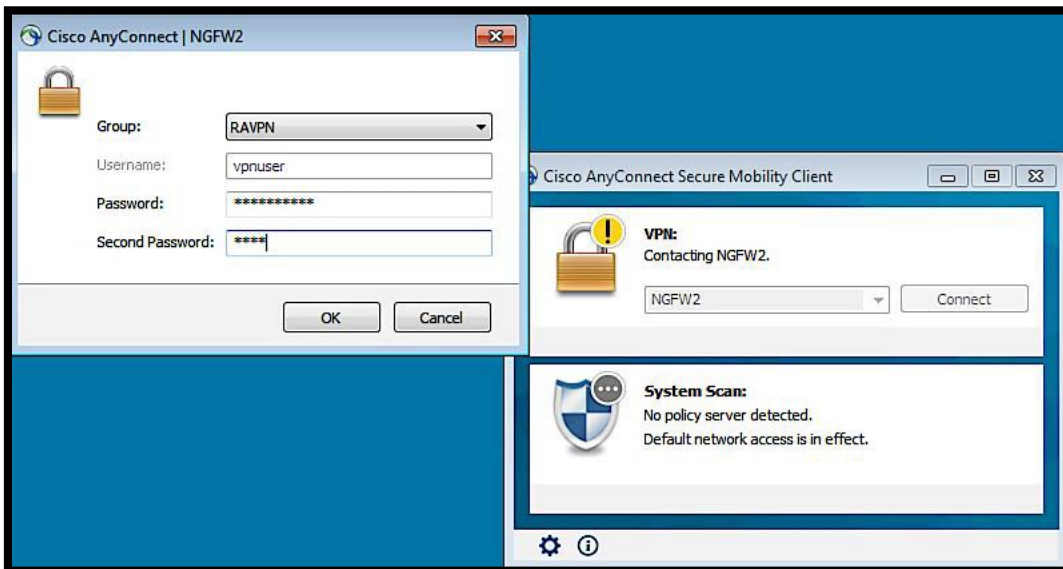
- c. If you are connected to the pod via VPN, use an RDP client on your laptop to connect to 198.18.133.23. Log in as Administrator, password C1sco12345.
2. Open AnyConnect from the Start Menu. You will see the Connect To field prepopulated with the NGFW2 FQDN. Click on Connect. Select the Group **RAVPN**.



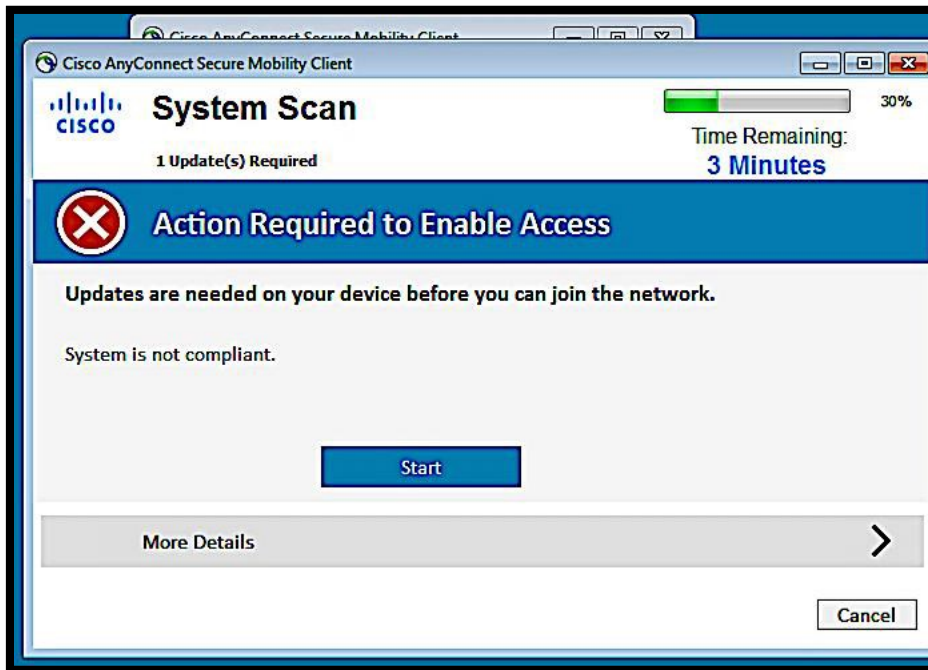
3. For testing we are going to use **vpnuser** and a client certificate of the same name. Select the vpnuser certificate.



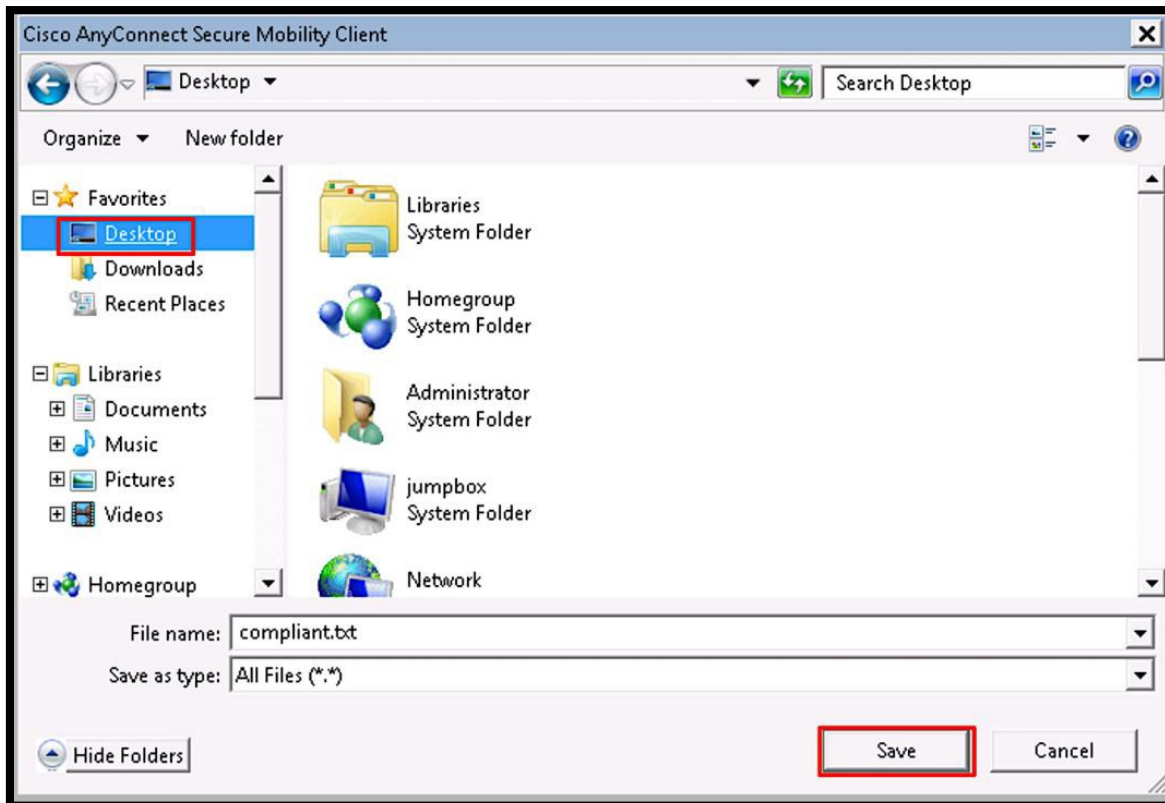
4. Enter the password as **C1sco12345**. If you configured Secondary Authentication, enter second password as **push**. Click on **OK**. Push is because we are going to use the Push method of Duo authentication, but this can be set to phone or SMS if you want. You should get a push notification on your mobile device to approve and when you do this, you will see the VPN tunnel getting established



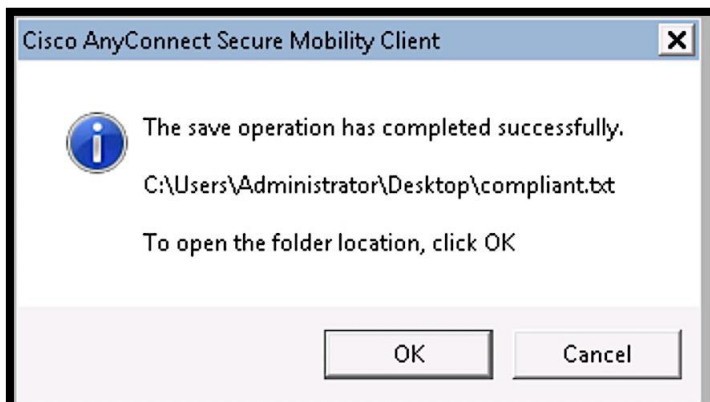
5. On the first connect, you will see the AnyConnect Compliance module being downloaded.
6. Since the system will not be compliant the first time we connect, you will be prompted to remediate. Click on Start.



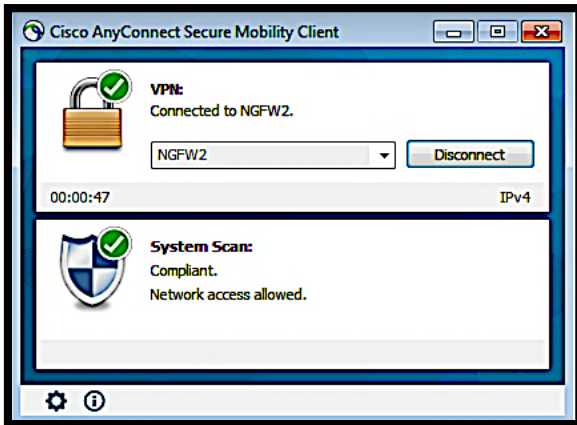
7. The module will prompt you save the compliant.txt file. Change the destination folder to the Desktop.



8. In the dialog box that says **The save operation has complete successfully**, you can click Cancel. There is no need to open the folder.



9. After the file is installed, the client will now declare your system compliant.



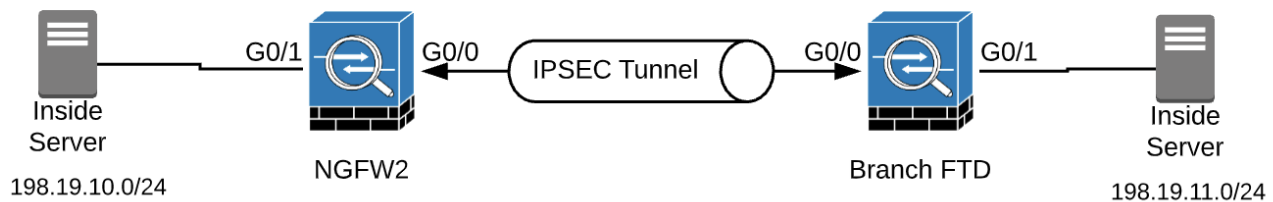
10. Using the bookmarks in the Firefox browser on Wkst2, confirm that you can now access the two internal websites that are bookmarked: Inside, Alt Inside.
11. In any one of these internal servers, click on the Files link, and then click on Zombies.pdf. This is considered malware. Confirm that it is blocked. Confirm that a benign file, like ProjectX.pdf, is not blocked

Scenario 2. Site-to-Site VPN Between FMC and FDM Managed Devices Using Certificate-Based Authentication

For this scenario we will be setting up certificate-based authentication for the Site to Site tunnel between NGFW2 (FDM) and NGFW Branch 1 (FMC). While configuring the tunnel also pay attention to the other enhancement which is available – the ability to explicitly configure dynamic peering on both FMC and FDM. This allows Firepower to scale its S2S VPN capabilities to accommodate branch management. The NGFW2 is protecting the local LAN network 198.19.10.0/24 and the Branch Office NGFW is protecting the local LAN network 198.19.11.0/24. At the end of this exercise a server behind NGFW2 should be able to access a server behind the Branch Office NGFW and vice-versa.

The objectives of this scenario are:

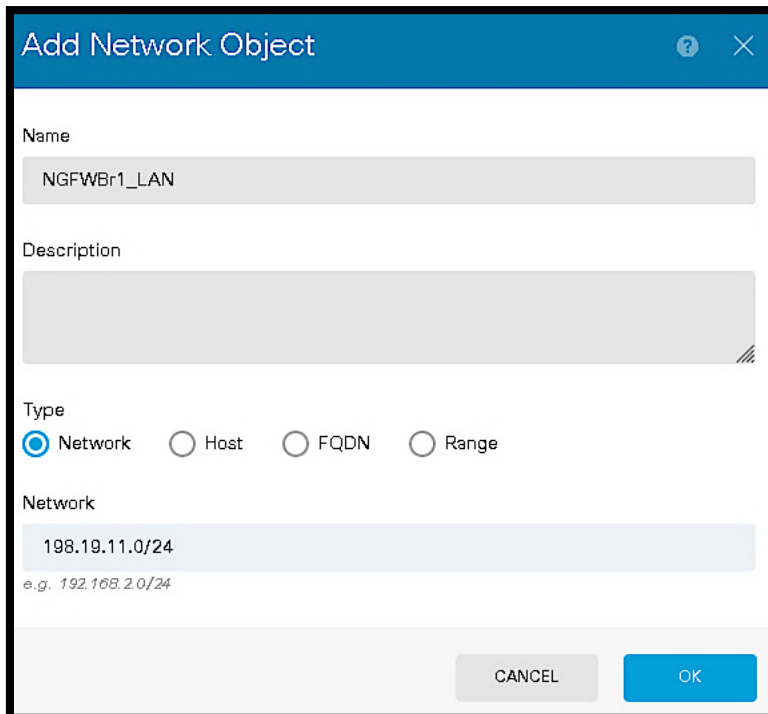
- Create the objects required for configuration of S2S VPN on NGFW2 and NGFW Branch 1.
- Run the S2S VPN setup wizard on NGFW2 and NGFW Branch 1.
- Test traffic through the S2S tunnel that's established.



FDM Pre-Configuration Steps

Creating Network Objects

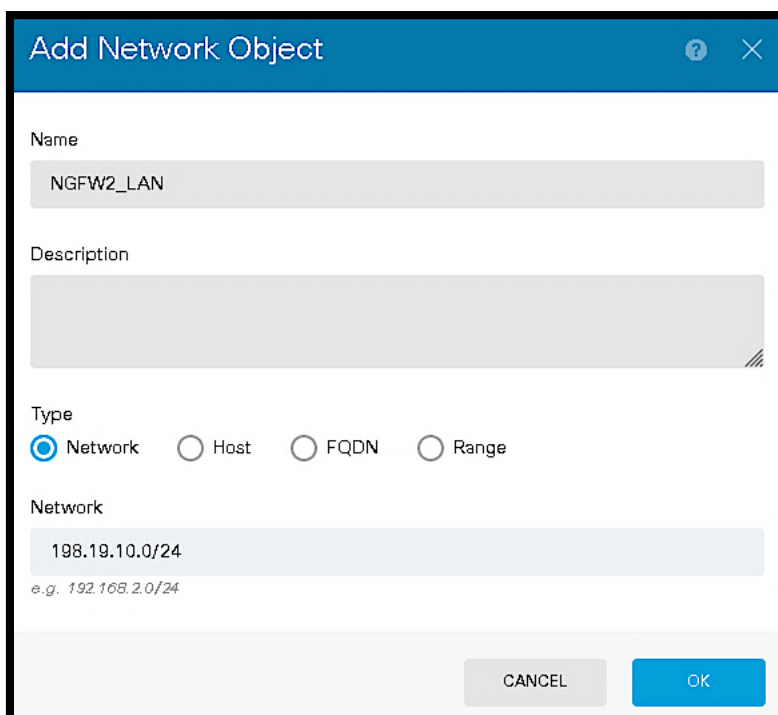
1. On the Jumpbox, open Firefox and click on **NGFW2 (FDM)** in the bookmarks list. Login to the NGFW2 FDM.
2. Navigate to **Objects > Networks**. Click on **+**.
 - a. Set Name to **NGFWBr1_LAN**, Type as **Network** and Network as **198.19.11.0/24**



The 'Add Network Object' dialog box is shown with the following details:

- Name:** NGFWBr1_LAN
- Description:** (Empty text area)
- Type:** Network (selected with a blue radio button), Host, FQDN, Range
- Network:** 198.19.11.0/24
- Example:** e.g. 192.168.2.0/24
- Buttons:** CANCEL, OK

- b. Repeat steps to add object with these details : Name as **NGFW2_LAN**, Type as **Network** and Network as **198.19.10.0/24**.



The 'Add Network Object' dialog box is shown with the following details:

- Name:** NGFW2_LAN
- Description:** (Empty text area)
- Type:** Network (selected with a blue radio button), Host, FQDN, Range
- Network:** 198.19.10.0/24
- Example:** e.g. 192.168.2.0/24
- Buttons:** CANCEL, OK

Upload Certificate for Site to Site VPN Certificated based authentication

NOTE: If you have done the RA VPN Scenario on FDM, skip Steps 1 and 2 in this Section.

1. Navigate to **Objects > Certificates**. Click on **+** and select **Add Trusted CA Certificate**.
 - a. Enter Name as **DCloud_CA**.
 - b. Click on **Upload Certificates**. Navigate to **Desktop > Certificates > Lab Certificates > Other Certificates** and select **dcloudCA.pem**
 - c. Click on **OK**.

Add Trusted CA Certificate

Name

DCloud_CA

Paste certificate, or choose file: **UPLOAD CERTIFICATE** dcloudCA.crt

```
-----BEGIN CERTIFICATE-----
MIIDfzCCAmegAwIBAgIJAMpXmP559bwYMA0GCSqGSIb3DQEBCwUAMFYxCzAJBgNV
BAYTAiVTMREwDwYDVQQHDAhTYW4gSm9zZTEWMBQGA1UECgwNQ2lzY28gU3lzdGVt
czEcMBoGA1UEAwwTaW5zaWRILmRjbG91ZC5sb2NhbDAeFw0xOTAzMTEyMTAxMDIa
Fw0yOTAzMDEyMTAxMDIaMFYxCzAJBgNVBAYTAiVTMREwDwYDVQQHDAhTYW4gSm9z
ZTEWMBQGA1UECgwNQ2lzY28gU3lzdGVtczEcMBoGA1UEAwwTaW5zaWRILmRjbG91
ZC5sb2NhbDCCASlwdDQYJKoZIhvcNAQEBBQADggEPADCCAQoCggEBAMr3f1+IE/kq
-----
```

CANCEL OK

2. Click on **+** and select **Add Internal Certificate**. Click on **Upload Certificate and Key**.
 - a. Enter Name as **NGFW2_Outside**.
 - b. Click on **Upload Certificates**. Navigate to **Desktop > Certificates > Lab Certificates** and select **ngfw-dcloud.crt**
 - c. Click on **Upload Key**. Navigate to **Desktop > Certificates > Lab Certificates** and select **ngfw-dcloud.key**
 - d. Click on **OK**.

Add Internal Certificate

Name
NGFW2_Outside

SERVER CERTIFICATE (USER AGENT)
Paste certificate, or choose file: UPLOAD CERTIFICATE ngfw-dcloud.crt

```

-----BEGIN CERTIFICATE-----
MIDzzCCAreAwIBAgIJAKdbAbuuBzPOMA0GCSqGSIb3DQEBCwUAMFYxCzAJBgNV
BAYTAiVTMREwDwYDVQQHDAhTYW4gSm9zZTEWMBQGA1UECgwNQ2l2Y28gU3lzdGVt
czEcMBoGA1UEAwwTaW5zaWRILmRjbG91ZC5sb2NhbDAeFw0xOTAzMTEyMTE5MDZa
Fw0yNDZzMDEyMTE5MDZaMFQxCzAJBgNVBAYTAiVTMREwDwYDVQQHDAhTYW4gSm9z

```

CERTIFICATE KEY
Paste key, or choose file: UPLOAD KEY ngfw-dcloud.key

```

-----BEGIN RSA PRIVATE KEY-----
MIIEpgIBAAKCAQEA0t3qL6tBMW7axEKvJ3pKtq1O2aIQ925IAN0DcJCsKGCocSTG
BsmDfQd8yB60em5/D5ymTh0C8Lu9HzjYkOVYnnwC3QSRwIQWRnhNRIOgcUSBr0u3
7v3uTFyRNIMgQtDnyzqdXxAC7D2mw4EsxhGGL5fWVKVWO4pPVFJHvrWxZpTBSLra
MVQNNIt4u5xMyQUTck7G67AZyUltqBhYFH90TEgwcYkYFIMTg7xWCNDc+k4p4Qg4

```

CANCEL OK

FDM Site to Site VPN Wizard

Device Summary
Site-to-Site VPN

Search

#	NAME	LOCAL INTERFACE	LOCAL NETWORKS	REMOTE NETWORKS	NAT EXEMPT	IKE V1	IKE V2	ACTIONS
There are no Site-to-Site connections yet. Start by creating the first Site-to-Site connection.								

CREATE SITE-TO-SITE CONNECTION

- Navigate to Device > **Site to Site VPN** > **View Configuration**. Click on **CREATE SITE-TO-SITE CONNECTION**.

Define Endpoints

In this page, we define the networks and endpoints involved in the Site to Site connection.

- Enter Connection Profile Name as **S2SVPN**

2. Select **outside** from Local VPN Access Interface dropdown.
3. Add **NGFW2_LAN** as the Local Network.
4. Enter Remote IP Address as **198.18.128.81**. This is IP address on the outside interface of the Branch NGFW.
5. Select **NGFWBr1_LAN** as the Remote Network.
6. Click on **Next**.

New Site-to-site VPN

1 Endpoint 2 Configuration 3 Summary

Define Endpoints

Identify the interface on this device, and the remote peer's interface IP address, that form the point-to-point VPN connection. Then, identify the local and remote networks that can use the connection. Traffic between these networks is protected using IPsec encryption.

Connection Profile Name: S2SVPN

LOCAL SITE

Local VPN Access Interface: outside (GigabitEthernet0/0)

Local Network: NGFW2_LAN

REMOTE SITE

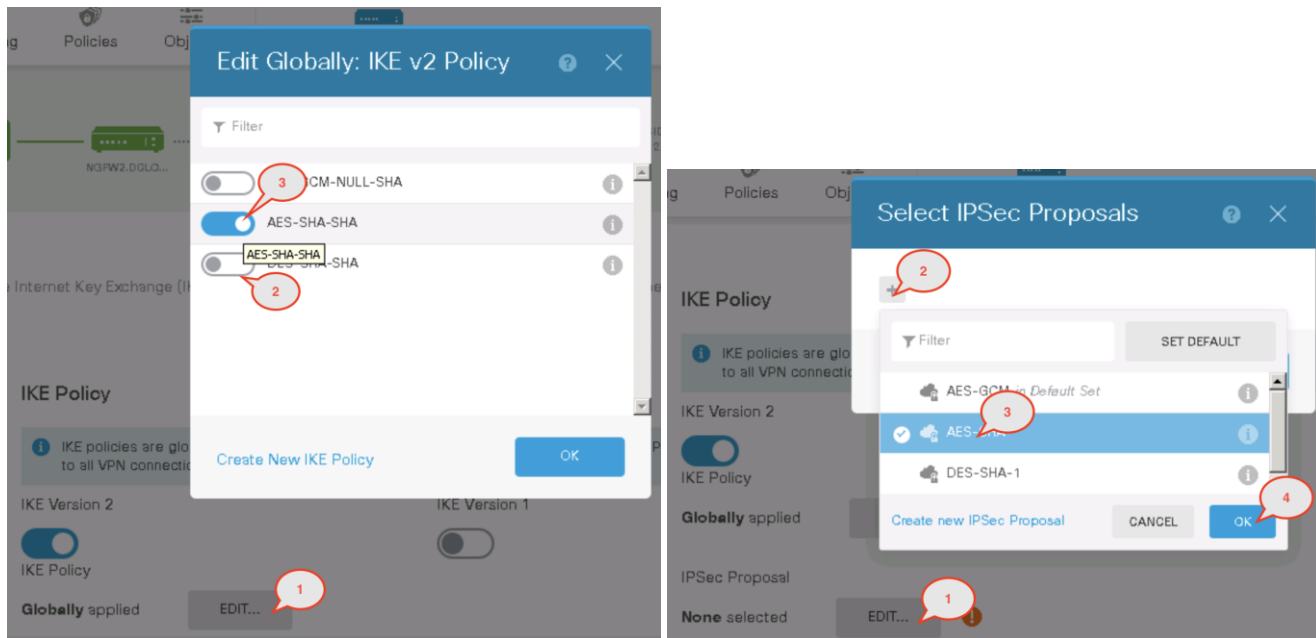
Remote IP Address: 198.18.128.81

Remote Network: NGFWBr1_LAN

CANCEL NEXT

Define Authentication and Privacy Settings for IKE and IPSEC.

1. Under the **IKE Policy**, **IKE version 1** is disabled and **IKE version 2** is Enabled. For this lab, we will continue with IKEv2.
2. Click **Edit** to configure the IKE security settings.
 - a. Disable **DES-SHA-SHA**
 - b. Enable **AES-SHA-SHA**
3. Click **Edit** to configure the IPSec proposal settings.
 - a. Click **+** icon to add from existing configuration.
 - b. Select **AES-SHA** and click **OK**
 - c. Click **OK**



NOTE: The user can define their own IKE and IPSEC Policies. However, for the purpose of this lab we are using pre-defined set of policies.

4. Under Authentication Type select **Certificate**. From the Certificate dropdown, select **NGFW2_Outside**.
5. Under Additional Options select in10 from the NAT Exempt dropdown.

Privacy Configuration

Select the Internet Key Exchange (IKE) policy and enter the pre-shared keys needed to authenticate the VPN connection. Then, select the IPsec proposals to use for encrypting traffic.

IKE Policy

i IKE policies are global, you cannot configure different policies per VPN. Any enabled IKE Policies are available to all VPN connections.

IKE Version 2



IKE Policy

Globally applied

EDIT...

IKE Version 1



IPSec Proposal

Custom set selected

EDIT...

Authentication Type



Pre-shared Manual Key



Certificate

Certificate

NGFW2_Outside

Additional Options

NAT Exempt

Inside (GigabitEthernet0/1)



Diffie-Hellman Group for Perfect Forward Secrecy

No Perfect Forward Secrecy (turned off)



BACK

NEXT

NOTE: The FDM adds in a NAT exemption statement for VPN traffic which is only visible via CLI.

6. Click on **Next**.

Review the Summary and click on Finish.

Click **Finish**.

S2SVPN Connection Profile

i Peer endpoint needs to be configured according to specified below configuration.

VPN Access Interface IP outside (198.18.133.82)

Peer IP Address 198.18.128.81



Network 198.19.10.0/24

Peer Network 198.19.11.0/24

IKE V2

IKE Policy aes,aes-192,aes-256-sha512,sha384,sha256,sha-sha512,sha384,sha256,sha-21,20,24,14,5

IPSec Proposal aes,aes-192,aes-256-sha-512,sha-384,sha-256,sha-1

Authentication Type Certificate

Certificate  NGFW2_Outside

IKE V1: DISABLED

OTHER

NAT Exempt Traffic from the protected network to the remote network is exempt from NAT when coming from the interface: inside.

Diffie-Hellman Group Null (not selected)

i Information is copied to the clipboard when you click Finish. You must allow the browser to access your clipboard for the copy to be succes...

BACK

FINISH

FDM Configuration to allow VPN Traffic at NGFW2

In this section, we will configure the Access-Control Policy at the NGFW2 to allow inbound traffic from the Branch FTD network.

1. Navigate to **Policies > Access Control**. Click on **+**.

- a. For the new rule, set Order to **1**.
- b. Name will be **S2SVPN_Access** and Action will be **Allow**.
- c. Set Source Zone to **outside_zone** and Source Networks to **NGFWBr1_LAN**

Add Access Rule

Order: 1 | Title: S2SVPN_Access | Action: Allow

Source/Destination | Applications | URLs | Users | Intrusion Policy | File policy | Logging

SOURCE			DESTINATION		
Zones	Networks	Ports	Zones	Networks	Ports/Protocols
outside_zone	NGFWBr1_LAN	ANY	ANY	ANY	ANY

- d. Navigate to **Intrusion Policy**. Enable Intrusion Policy and set the Level to **Balanced Security and Connectivity**

Add Access Rule

Order: 1 | Title: S2SVPN_Access | Action: Allow

Source/Destination | Applications | URLs | Users | **Intrusion Policy** | File policy | Logging

INTRUSION POLICY ☒

Level of Intrusion Policy: **Balanced Security and Connectivity**

Balanced Security and Connectivity

This policy is designed to balance overall network performance with network infrastructure security. This policy is appropriate for most networks. Select this policy for most situations where you want to apply intrusion prevention.

PREVENTING INTRUSIONS

Use intrusion policies as a last line of defense against unwanted traffic that you are otherwise allowing. An intrusion policy examines decoded packets for intrusions, exploits, and other attacks based on patterns, and can block or alter malicious traffic. Cisco delivers several intrusion policies with the Firepower system. These policies are designed by the Cisco Talos Security Intelligence and Research Group, who set the intrusion and preprocessor rule states and advanced settings.

- e. Navigate to **File Policy**. Select **Block Malware All**.

Add Access Rule

Order	Title	Action
1	S2SVPN_Access	Allow

Source/Destination Applications URLs Users Intrusion Policy **File policy** Logging

SELECT THE FILE POLICY

Block Malware All

CONTROLLING FILES AND MALWARE

Use file policies to detect malicious software, or malware, using Advanced Malware Protection for Firepower (AMP for Firepower.) You can also use file policies to perform file control, which allows control over all files of a specific type regardless of whether the files contain malware

Query the AMP cloud to determine if files traversing your network contain malware, then block files that represent threats.

- f. Navigate to **Logging**. Select **At End of Connection** for Select **Log Action**.
- g. Click on **OK**.

FMC Pre-Configuration Steps

Now, we will perform the configuration steps on the FMC for the Branch FTD.

Creating Network Objects

1. On the Jumpbox, open Firefox and click on FMC in the bookmarks list. Login to the FMC.
2. Navigate to **Objects > Object Management > Network**. From the **Add Network** dropdown, click on **Add Object**.
 - a. Enter **NGFWBr1_LAN** as the Name.
 - b. Set Network to **Network** and type in **198.19.11.0/24**.
 - c. Click on **Save**.
3. Click on **Add Object**.
 - a. Enter **NGFW2_LAN** as the Name.
 - b. Set Network to **Network** and type in **198.19.10.0/24**.
 - c. Click on **Save**

New Network Object

Name

NGFWBr1_LAN

Description

Network

☐ Host
☐ Range
☒ Network
☐ FQDN

198.19.11.0/24

☐ Allow Overrides

Cancel

Save

New Network Object

Name

NGFW2_LAN

Description

Network

☐ Host
☐ Range
☒ Network
☐ FQDN

198.19.10.0/24

☐ Allow Overrides

Cancel

Save

Upload Certificate for Site to Site VPN Certificated based authentication

1. Navigate to **Objects > PKI > Cert Enrollment**. Click on **Add Cert Enrollment**.
 - a. Enter Name as **NGFWBr1_Outside**.
 - b. Set Enrollment Type as **PKCS 12 File**.
 - c. Click on **Browse PKCS12 File** and navigate to **Desktop > Certificates > Lab Certificates > Other Certificates**. Select **ngfwbr1-outside.pfx**.
 - d. Enter Passphrase as **C1sco12345**.
 - e. Click on **Save**

Add Cert Enrollment

Name*

NGFWBr1_Outside

Description

CA Information

Certificate Parameters

Key

Revocation

Enrollment Type:

PKCS12 File

PKCS12 File*

ngfwbr1-outside.pfx

Browse PKCS12 File

Passphrase:

☐ Allow Overrides

Cancel

Save

2. To enroll NGFW Branch 1 with the certificate we just uploaded, navigate to **Devices > Certificates**. Click on **Add**.
 - a. Select **NGFWBR1** for Device.
 - b. Select **NGFWBr1_Outside** for Cert Enrollment.
 - c. Click on **Add**.

Add New Certificate

Add a new certificate to the device using cert enrollment object which is used to generate CA and identify certificate.

Device*: NGFWBR1

Cert Enrollment*: NGFWBr1_Outside

Cert Enrollment Details:

Name: NGFWBr1_Outside
Enrollment Type: PKCS12 file
SCEP URL: NA

Cancel Add

- d. The certificate will be immediately enrolled on the device and should be successful

▼ NGFWBR1				
NGFWBr1_Outside	Global	PKCS12 file	CA ID	

FMC Site to Site VPN Wizard

Here only one of the FTDs is managed by the FMC so the wizard can prepopulate the device information for only NGFWBR1. For the other endpoint, there is a provision in the Wizard for externally managed devices. There are three kinds of VPN topologies available in the FMC Site to Site VPN wizard:

- Point to Point – This is the deployment method of choice where we need to establish a VPN tunnel between two endpoints.
- Hub and Spoke – This is the deployment method of choice for a use case involving one HQ/Main office and multiple Remote/Branch locations. It supports adding Spokes as **Extranet** if the Remote VPN endpoints are not managed by FMC.
- Full Mesh – This is the deployment method of choice when multiple VPN endpoints have tunnel connectivity to each other.

Here we are going to be using Point to Point.

1. Navigate to **Devices > VPN > Site to Site**. Click on **Add VPN** then click **Firepower Threat Defense Device**
 - a. Enter Topology Name as **S2SVPN**
 - b. Confirm that for Network Topology, **Point to Point** is selected. Confirm that for IKE Version, **IKEv1** is **not checked**, and **IKEv2** is **checked**.
 - c. Under Endpoints, click on + for **Node A**.
 - i. Select **Extranet** as Device.
 - ii. Enter **NGFW2** as Device Name.

- iii. Set IP Address as **Static** and **198.18.133.82**.
- iv. Under Protected Networks, add the **NGFW2_LAN** object.
- v. Click on **OK**.

The screenshot shows the 'Create New VPN Topology' window on the left and the 'Add Endpoint' window on the right. The 'Create New VPN Topology' window has the following fields and options:

- Topology Name: S2SVPN (1)
- Network Topology: Point to Point (2)
- IKE Version: IKEv1 (3)
- Endpoints: IKE, IPsec, Advanced
- Node A: Table with columns Device Name, VPN Interface, Protected Networks
- Node B: Table with columns Device Name, VPN Interface, Protected Networks

The 'Add Endpoint' window has the following fields and options:

- Device: NGFW2 (4)
- Device Name: NGFW2 (5)
- IP Address: Static (6), 198.18.133.82 (7)
- Certificate Map: (8)
- Protected Networks: (9)
- Subnet / IP Address (Network) (10)
- Cancel, OK (11)

The 'Network Objects' window is also visible, showing the 'Available Networks' list with 'NGFW2_LAN' (8) and 'NGFW2_LAN' (9) in the 'Selected Networks' list. The 'NGFW2_LAN' object has the IP address 198.19.10.0/24. The 'Cancel' and 'OK' buttons are at the bottom right (10).

- d. Under Endpoints, click on **+** for **Node B**.
 - i. Select **NGFWBR1** as Device.
 - ii. Set Interface as **outside**.
 - iii. Set IP Address as **198.18.128.81**.
 - iv. Under Protected Networks, add the **NGFWBr1_LAN** object.
 - v. Click on **OK**.

The screenshot shows the 'Create New VPN Topology' window on the left and the 'Add Endpoint' window on the right. The 'Create New VPN Topology' window has the following fields and options:

- Topology Name: S2SVPN
- Network Topology: Point to Point, Hub and Spoke, Full Mesh
- IKE Version: IKEv1 (checked), IKEv2
- Endpoints: IKE, IPsec, Advanced
- Node A: Table with columns Device Name, VPN Interface, Protected Networks
- Node B: Table with columns Device Name, VPN Interface, Protected Networks

The 'Add Endpoint' window has the following fields and options:

- Device: NGFWBR1 (1)
- Interface: outside (2)
- IP Address: 198.18.128.81 (3)
- Connection Type: Bidirectional (4)
- Certificate Map: (5)
- Protected Networks: (6)
- Subnet / IP Address (Network) (7)
- Access List (Extended) (8)
- Cancel, OK (9)

The 'Network Objects' window is also visible, showing the 'Available Networks' list with 'NGFWBr1_LAN' (6) and 'NGFWBr1_LAN' (7) in the 'Selected Networks' list. The 'NGFWBr1_LAN' object has the IP address 198.19.11.0/24. The 'Cancel' and 'OK' buttons are at the bottom right (8).

Topology Name:*

Network Topology:
☒ Point to Point ☐ Hub and Spoke ☐ Full Mesh

IKE Version:* ☐ IKEv1 ☒ IKEv2

Endpoints ☒ IKE ☐ IPsec ☐ Advanced

Node A: +

Device Name	VPN Interface	Protected Networks	
NGFW2	198.18.133.82	NGFW2_LAN	

Node B: +

Device Name	VPN Interface	Protected Networks	
NGFWBr1	outside/198.18.128.81	NGFWBr1_LAN	

Ensure the protected networks are allowed by access control policy of each device.

[How To](#) [Cancel](#) [Save](#)

2. Select the **IKE** tab.
 - a. Under IKEv2 Settings, for Policy, select **AES-SHA-SHA**.
 - b. Under IKEv2 Settings, for Authentication Type, select Certificate.
 - c. Select **NGFWBr1_Outside** for Certificate.

Topology Name:*
S2SVPN

Network Topology:
☒ Point to Point
 ☐ Hub and Spoke
 ☐ Full Mesh

IKE Version:* ☐ IKEv1 ☒ IKEv2

Endpoints IKE IPsec Advanced

IKEv1 Settings

Policy:* preshared_sha_aes256_dh5_5 +

Authentication Type: Pre-shared Automatic Key

Pre-shared Key Length:* 24 Characters (Range 1-127)

IKEv2 Settings

Policy:* AES-SHA-SHA +

Authentication Type: Certificate

Certificate:* NGFWBr1_Outside +

Cancel Save

3. Select the **IPsec** tab, change the IKEv2 IPsec Proposal to **AES-SHA**.

- Click on the **pencil** (Edit) icon by IKEv2 IPsec Proposal.
- Click on **AES-SHA** and Click **Add**.
- Delete **AES-GCM**.
- Uncheck **Enable Reverse Route Injection**

Topology Name:* S2SVPN

Network Topology: ☒ Point to Point ☐ Hub and Spoke ☐ Full Mesh

IKE Version:* ☐ IKEv1 ☒ IKEv2

Endpoints IKE **IPsec** Advanced

Transform Sets: IKEv1 IPsec Proposals / IKEv2 IPsec Proposals 

tunnel_aes256_sha AES-GCM

☐ Enable Security Association (SA) Strength Enforcement

☒ Enable Reverse Route Injection

☐ Enable Perfect Forward Secrecy

Modulus Group: 14

Lifetime Duration*: 28800 Seconds (Range 120-2147483647)

Lifetime Size: 4608000 Kbytes (Range 10-2147483647)

ESPv3 Settings [How To](#)

Deploy  admin

IKEv2 IPsec Proposal

Available Transform Sets

Q Search

AES-GCM

AES-SHA

DES_SHA-1

Selected Transform Sets

Add AES-SHA

Cancel OK

NOTE: The user can define their own IKE and IPSEC Policies. However, for the purpose of this lab we are using pre-defined set of policies. You can view the details of these policies under the Object Management page at **Objects > VPN**. Also, there are more settings available under the **Advanced** tab such as Keepalives, fragmentation, NAT-T etc. For the purpose of this lab, we are using default configuration of those settings.

e. Click **OK**

4. Click **Save**.

FMC Configuration to Allow VPN Traffic

1. Navigate to **Devices > NAT** and Edit **Branch NAT Policy**

2. Click **Add Rule**.

a. Select **InZone** for Source Interface Objects and **OutZone** for Destination Interface Objects

The screenshot shows the 'Add NAT Rule' configuration window. The 'Interface Objects' tab is active. In the 'Available Interface Objects' list, 'InZone' and 'OutZone' are visible. 'InZone' is added to the 'Source Interface Objects' list, and 'OutZone' is added to the 'Destination Interface Objects' list. The 'In Category' and 'NAT Rules Before' dropdowns are highlighted with a red box. The 'Enable' checkbox is checked. The 'Description' field is empty. The 'Cancel' and 'OK' buttons are at the bottom right.

b. Under the **Translation** tab, select **NGFWBr1_LAN** as Original Source and Translated Source. Select **NGFW2_LAN** as the Original Destination and Translated Destination.

c. On the **Advanced** Tab select **Do not proxy ARP on Destination Interface**

Add NAT Rule

NAT Rule: Manual NAT Rule

Insert: In Category NAT Rules Before

Type: Static

☒ Enable

Description:

Interface Objects **Translation** PAT Pool Advanced

Original Packet

Original Source: NGFWBr1_LAN

Original Destination: Address

Original Source Port:

Original Destination Port:

Translated Packet

Translated Source: Address

Translated Destination: NGFWBr1_LAN

Translated Source Port:

Translated Destination Port:

Cancel OK

Add NAT Rule

NAT Rule: Manual NAT Rule

Insert: In Category NAT Rules Before

Type: Static

☒ Enable

Description:

Interface Objects Translation **Advanced** PAT Pool

☐ Translate DNS replies that match this rule

☐ Falthrough to Interface PAT(Destination Interface)

☐ IPv6

☐ Net to Net Mapping

☒ Do not proxy ARP on Destination Interface

☐ Perform Route Lookup for Destination Interface

☐ Unidirectional

Cancel OK

- d. Click **OK** to save the NAT rule. Click on **Save**

NOTE: Enabling Do not proxy ARP on Destination Interface is critical in this lab exercise. If you miss this step, the NGFWBR1 will start replying to ARP requests received on the outside interface, for IP addresses which are part of the NGFWBR1 outside interface subnet. This might cause your pod to have access issues, since all devices are managed in band.

3. Navigate to **Policies > Access Control** and edit **Branch Access Control Policy**. Click on **Add Rule**.
 - a. Enter Name as **S2SVPN_Access**.
 - b. **Change Insert to Into Mandatory**.

- c. Add **OutZone** to **Source Zone** and **NGFW2_LAN** as **Source Networks**.
- d. Under **Inspection**, set Intrusion Policy as **Demo Intrusion Policy** and File Policy as **Demo File Policy**.
- e. Under **Logging**, check **Log at End of Connection**.
- f. Click on **Add**. Click on **Save**.

Branch Access Contro

Enter Description

You have unsaved changes [Analyze Hit Counts](#) [Save](#) [Cancel](#)

[Inheritance Settings](#) | [Policy Assignments \(1\)](#)

Prefilter Policy: [Default Prefilter Policy](#) SSL Policy: [None](#) Identity Policy: [None](#)

Rules Security Intelligence HTTP Responses Logging Advanced

[Filter by Device](#) ☐ Show Rule Conflicts [+ Add Category](#) [+ Add Rule](#)

#	Name	Source Zones	Dest Zones	Source Networks	Dest Networks	VLAN Tags	Users	Application	Source Ports	Dest Ports	URLs	Source SGT	Dest SGT	Action	Icons
▼ Mandatory - Branch Access Control Policy (1-1)															
1	S2SVPN...	OutZone	Any	NGFW2...	Any	Any	Any	Any	Any	Any	Any	Any	Any	Allow	Icons

Deploy and test the configuration

Deploy and verify the configuration

1. Deploy the configurations on both NGFWBR1 on the FMC and the NGFW2 FDM.
2. Click **Proceed**. Wait for the deployment to complete. You can monitor the status of the deployment by clicking on the **Notifications** tab.

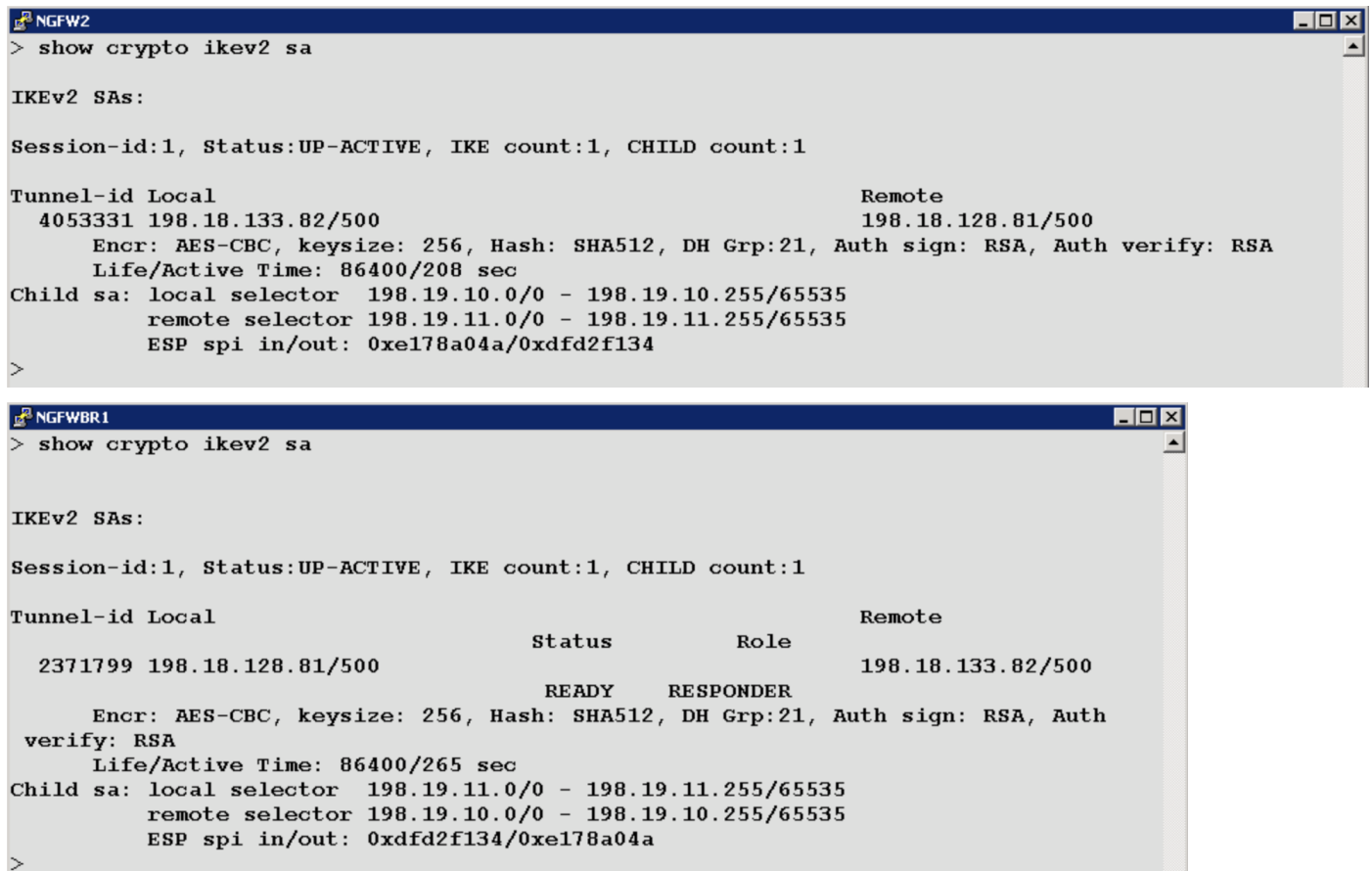
NOTE: You will get a warning during deployment. The deployment warning is due to the use of DH groups 5 and 24 in the IKEv2 Policy. DH Groups 2,5 and 24 have been deprecated in version 6.5 and in later versions they will not be supported.

Test the Configuration

1. From jumpbox, open PUTTY and login to NGFW2 and NGFWBR1 with credentials – User: **admin** Password: **C1sco12345**
2. From the **NGFW2** CLI, type **show crypto ipsec sa peer 198.18.128.81**. There should be no IPSec security associations.
3. From the **NGFWBR1** CLI, type **show crypto ipsec sa peer 198.18.133.82**. There should be IPSec security associations.
4. On the Jumpbox, open a PUTTY session to the Inside Linux Server. Login as root/C1sco12345.
5. Verify the routing on this server using the command **route -n**. There should be route for 198.19.11.0/24 via 198.19.10.2. If the route does not exist, add it with the command **route add -net 198.19.11.0 netmask 255.255.255.0 gw 198.19.10.2**
 - a. This will set the path for the Inside Linux server to reach the Branch1 LAN network via the NGFW2.
6. The Inside Linux Server sits in the LAN behind NGFW2. We will ping a workstation in Branch 1 with an IP address of 198.19.11.225. The ping should be successful. (The first one might drop to initiate the tunnel). On the Inside Linux Server, run the command **ping 198.19.11.225**.

```
[root@inside ~]# ping 198.19.11.225
PING 198.19.11.225 (198.19.11.225) 56(84) bytes of data.
64 bytes from 198.19.11.225: icmp_seq=2 ttl=128 time=2.91 ms
64 bytes from 198.19.11.225: icmp_seq=3 ttl=128 time=54.0 ms
64 bytes from 198.19.11.225: icmp_seq=4 ttl=128 time=2.85 ms
64 bytes from 198.19.11.225: icmp_seq=5 ttl=128 time=2.25 ms
64 bytes from 198.19.11.225: icmp_seq=6 ttl=128 time=2.88 ms
64 bytes from 198.19.11.225: icmp_seq=7 ttl=128 time=3.12 ms
64 bytes from 198.19.11.225: icmp_seq=8 ttl=128 time=2.10 ms
^C
--- 198.19.11.225 ping statistics ---
8 packets transmitted, 7 received, 12% packet loss, time 7008ms
rtt min/avg/max/mdev = 2.109/10.027/54.046/17.973 ms
[root@inside ~]#
```

7. You can see the tunnel status by logging into either NGFW2 or NGFWBR1 via Putty and executing **show crypto ikev2 sa** and **show crypto ipsec sa**.



The first screenshot shows the terminal for NGFW2. The command `show crypto ikev2 sa` is executed, displaying the status of the IKEv2 Security Association (SA). The output shows a single SA with Session-id:1, Status:UP-ACTIVE, IKE count:1, and CHILD count:1. The tunnel is identified as Local 4053331 and Remote 198.18.128.81/500. The encryption is AES-CBC with a key size of 256, and the hash is SHA512. The authentication method is RSA. The child SA is local selector 198.19.10.0/0 - 198.19.10.255/65535 and remote selector 198.19.11.0/0 - 198.19.11.255/65535. The ESP spi in/out is 0xe178a04a/0xdfd2f134.

The second screenshot shows the terminal for NGFWBR1. The command `show crypto ikev2 sa` is executed, displaying the status of the IKEv2 Security Association (SA). The output shows a single SA with Session-id:1, Status:UP-ACTIVE, IKE count:1, and CHILD count:1. The tunnel is identified as Local 2371799 and Remote 198.18.133.82/500. The encryption is AES-CBC with a key size of 256, and the hash is SHA512. The authentication method is RSA. The child SA is local selector 198.19.11.0/0 - 198.19.11.255/65535 and remote selector 198.19.10.0/0 - 198.19.10.255/65535. The ESP spi in/out is 0xdfd2f134/0xe178a04a.

8. You can verify the tunnel authentication method that is being used by logging into either NGFW2 or NGFWBR1 via Putty and executing **show vpn-sessiondb detail I2I**.

NGFW01

> show vpn-sessiondb detail 121

Session Type: LAN-to-LAN Detailed

```

Connection : 198.18.133.82
Index      : 1                      IP Addr   : 198.18.133.82
Protocol   : IKEv2 IPsec
Encryption : IKEv2: (1)AES256 IPsec: (1)AES256
Hashing    : IKEv2: (1)SHA512 IPsec: (1)SHA512
Bytes Tx   : 168                      Bytes Rx  : 1092
Login Time : 04:20:36 UTC Fri Apr 3 2020
Duration   : 0h:05m:44s
Tunnel Zone : 0

```

IKEv2 Tunnels: 1

IPsec Tunnels: 1

IKEv2:

```

Tunnel ID : 1.1
UDP Src Port : 500                      UDP Dst Port : 500
Rem Auth Mode: rsaCertificate
Loc Auth Mode: rsaCertificate
Encryption : AES256                      Hashing      : SHA512
Rekey Int (T): 86400 Seconds             Rekey Left(T): 86056 Seconds
PRF        : SHA512                      D/H Group   : 21
Filter Name :

```

IPsec:

```

Tunnel ID : 1.2
Local Addr : 198.19.11.0/255.255.255.0/0/0
Remote Addr : 198.19.10.0/255.255.255.0/0/0
Encryption : AES256                      Hashing      : SHA512
Encapsulation: Tunnel
Rekey Int (T): 28800 Seconds             Rekey Left(T): 28456 Seconds
Rekey Int (D): 4608000 K-Bytes           Rekey Left(D): 4607999 K-Bytes
Idle Time Out: 30 Minutes                 Idle TO Left : 24 Minutes
Conn Time Out: 2068858 Minutes           Conn TO Left : 2068852 Minutes
Bytes Tx     : 168                      Bytes Rx     : 1092
Pkts Tx      : 2                      Pkts Rx      : 13

```

NGFW2

> show vpn-sessiondb detail 121

Session Type: LAN-to-LAN Detailed

```

Connection : 198.18.128.81
Index      : 1                      IP Addr   : 198.18.128.81
Protocol   : IKEv2 IPsec
Encryption : IKEv2: (1)AES256 IPsec: (1)AES256
Hashing    : IKEv2: (1)SHA512 IPsec: (1)SHA512
Bytes Tx   : 1092                      Bytes Rx  : 168
Login Time : 04:20:36 UTC Fri Apr 3 2020
Duration   : 0h:06m:31s
Tunnel Zone : 0

```

IKEv2 Tunnels: 1

IPsec Tunnels: 1

IKEv2:

```

Tunnel ID : 1.1
UDP Src Port : 500                      UDP Dst Port : 500
Rem Auth Mode: rsaCertificate
Loc Auth Mode: rsaCertificate
Encryption : AES256                      Hashing      : SHA512
Rekey Int (T): 86400 Seconds             Rekey Left(T): 86009 Seconds
PRF        : SHA512                      D/H Group   : 21
Filter Name :

```

IPsec:

```

Tunnel ID : 1.2
Local Addr : 198.19.10.0/255.255.255.0/0/0
Remote Addr : 198.19.11.0/255.255.255.0/0/0
Encryption : AES256                      Hashing      : SHA512
Encapsulation: Tunnel
Rekey Int (T): 28800 Seconds             Rekey Left(T): 28409 Seconds
Rekey Int (D): 4608000 K-Bytes           Rekey Left(D): 4607999 K-Bytes
Idle Time Out: 30 Minutes                 Idle TO Left : 23 Minutes
Conn Time Out: 2073529 Minutes           Conn TO Left : 2073522 Minutes
Bytes Tx     : 1092                      Bytes Rx     : 168
Pkts Tx      : 13                      Pkts Rx      : 2

```

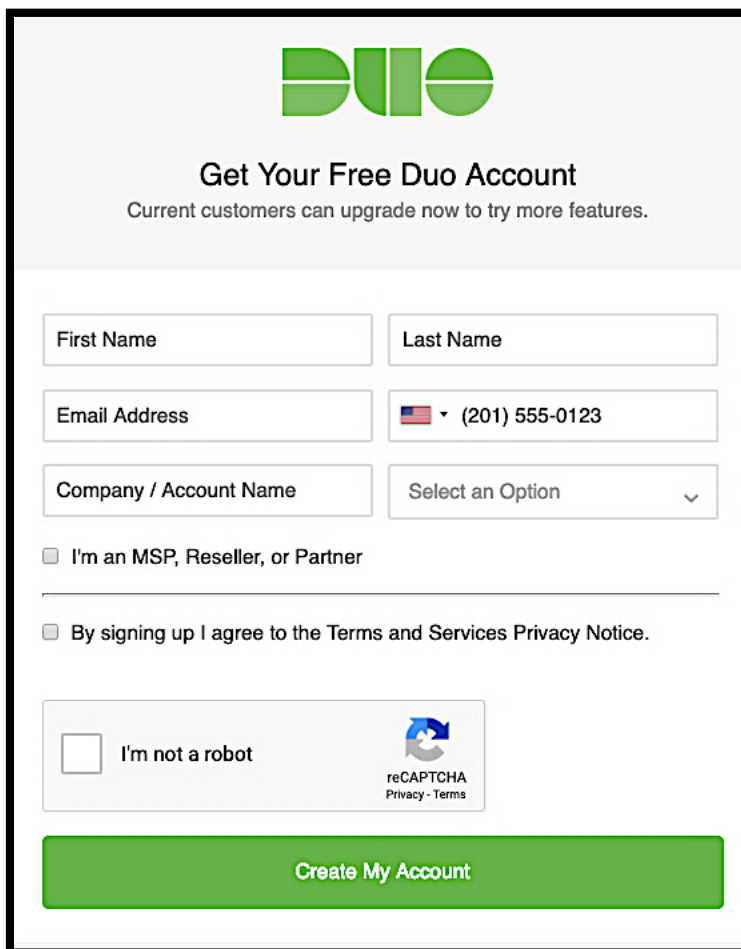
NOTE: In order to save time, ISE has been pre-configured with all required configuration for all of the lab exercises. If you want to inspect the ISE configuration, see Appendix D. Enabling **Do not proxy ARP on Destination Interface** is critical in this lab exercise. If you miss this step, your pod may have access issues, since all devices are managed in band.

Appendix A. DUO Account Setup

Create Duo Account and enroll your mobile device

NOTE: You can skip this section and continue to the Add Firepower application and enroll VPN user to Duo Account section if you already have a Duo Account and have your mobile device registered to this account.

2. Since we will be using Duo as a factor in our RA VPN authentication, let's first setup a Duo Account for ourselves. Duo allows you to create a free trial account and link it to your mobile device. This will be sufficient for our lab exercise.
3. Navigate to <https://duo.com/docs/getting-started> and follow steps 1-4 under the **Getting Started** section.



The screenshot shows the Duo account creation page. At the top is the Duo logo. Below it is the heading "Get Your Free Duo Account" and a subtext "Current customers can upgrade now to try more features." The form contains several input fields: "First Name", "Last Name", "Email Address", and a phone number field with a dropdown for country (USA) and a pre-filled number "(201) 555-0123". There is also a "Company / Account Name" field and a "Select an Option" dropdown. Below these fields are two checkboxes: "I'm an MSP, Reseller, or Partner" and "By signing up I agree to the Terms and Services Privacy Notice." At the bottom left is a checkbox labeled "I'm not a robot" next to a reCAPTCHA widget. A large green "Create My Account" button is at the bottom.

1. Once you are done, we can proceed to the next section to add our test VPN user and the Firepower application to your account.

Add Firepower application and enroll VPN user to Duo Account

- Log in to the **Duo Admin Panel** [https://admin.duosecurity.com] and navigate to **Applications**.
- Locate **Cisco Firepower Threat Defense VPN** in the applications list

Dashboard > Applications > Protect an Application

Protect an Application

Add an application that you'd like to protect with Duo two-factor authentication.
You can start with a small "proof-of-concept" installation — it takes just a few minutes, and you're the only one that will see it, until you decide to add others.

Documentation: [Getting Started](#)

Choose an application below to get started.

**Cisco Firepower Threat Defense VPN**[Protect this Application](#) | [Read the documentation](#)

- Click **Protect this Application** to get your integration key, secret key, and API hostname

Dashboard > Applications > Cisco Firepower Threat Defense VPN

Cisco Firepower Threat Defense VPN

Authentication Log |  Remove Application

Follow the [Cisco Firepower Threat Defense \(FTD\) Remote Access VPN application instructions](#).

[Reset Secret Key](#)

Details

Integration key	DIBS6UGGW1KH44Y7LJCH	select
Secret key	Click to view.	select
Don't write down your secret key or share it with anyone.		
API hostname	api-29acc416.duosecurity.com	select

NOTE: Treat your secret key like a password. The security of your Duo application is tied to the security of your secret key (skey). Secure it as you would any sensitive credential. Don't share it with unauthorized individuals or email it to anyone under any circumstances

7. There is a lot more functionality associated with the application such as Policy, but this is out of the scope of the lab.
8. Let's now add our VPN user account that will be used for testing. Navigate to **Users** on the **Duo Admin Panel** and click on **Add User**. Type in **vpnuser** for the **Username** field and click on **Add User**.

Dashboard > Users > Add User

Add User

Adding Users
Most applications allow users to enroll themselves after they complete primary authentication.
[Learn more about adding users](#)

Username
Should match the primary authentication username.

Add User

NOTE: If you cannot receive SMS messages for enrollment, follow step #6, otherwise follow from step #7 onwards.

9. In the **vpnuser** dashboard add in an email address that you can use for enrollment. Click on **Save Changes**. Now you can click on the **Send Enrollment Email** link to get a message that contains a link that you can use to add a phone or other 2FA authentication device. Jump to **step #12**.

Dashboard > Users > vpnuser

vpnuser

Logs | Send Enrollment Email | Send to Trash

This user has not enrolled yet. See our [enrollment documentation](#) to learn more about enrolling users.

10. In the **vpnuser** dashboard add in your phone number so that an enrollment message can be sent. Scroll down on the new user's details page to the **Phones** table and click on **Add Phone**

Phones

To use two-factor with a specific phone, type 'push1', 'sms1', 'phone2', 'sms2', etc, as your factor. You may rearrange the phones by dragging and dropping in the table.

[Add Phone](#)

This user has no phones. [Add one.](#)

- Chose Phone, and type in the phone number (leave this field blank if adding a tablet). Click the Add Phone button.

[Dashboard](#) > [Users](#) > [avandalay](#) > Add Phone

Add Phone

Type

☒

Phone

☐

Tablet

Phone number



+1 734-555-8008

[Show extension field](#)

[Add Phone](#)

- Choose the appropriate phone Type and Platform from the drop-down menus and enter a Device name (this field can be left blank). If you know the device is a smartphone but aren't sure exactly what the platform is, choose Generic Smartphone and the actual platform will be set when the user completes activation. Click the Save Changes button.

Settings

Number	 +1 734-555-8008	Show extension settings
Device name		
	Optional. Examples: "Work phone", "Old iPod touch"	
Type	Mobile	
Platform	iOS	
Save Changes		

- Click the Activate Duo Mobile link in the Device Info section. This link is only available when you set the phone type to Mobile and selected something other than Unknown as the platform. Then on the next page click the Generate Duo Mobile Activation Code button.



Activate Duo Mobile

This form allows you to generate a new activation code for this phone's Duo Mobile application. The Duo Mobile application allows the user to generate passcodes on their mobile device or authenticate via Duo Push.

Note: Generating an activation code will invalidate any existing Duo Mobile credentials for this device until it is activated with the new activation code.

Phone 734-555-4211

Expiration 24 hours after generation

Generate Duo Mobile Activation Code

14. Next, you'll see two text messages that you can send. The first has a link that helps the user install Duo Mobile. The second message has a code that the user can use to immediately add the account to their Duo Mobile app. Click the Send Instructions by SMS button to send the text messages to the user's phone.

Activate Duo Mobile

A new Duo Mobile activation code has been generated, and any old credentials have been invalidated. You now have the option to deliver the installation instructions and/or activation instructions to the user by SMS.

Phone 734-555-4211

Installation Instructions ☒ Send installation instructions via SMS

Welcome to Duo! Please install Duo Mobile from your app store.

Activation Instructions ☒ Send activation instructions via SMS

To activate the app, tap and open this link with Duo Mobile: <https://m-xxxxxxxxx.duosecurity.com/activate/LSdDPKMB2312HktaAKcll>

Send Instructions by SMS or skip this step

15. You would have received an email message or an SMS message with an activation link and QR code, as well as links to the Duo Mobile app for all supported platforms. You need to either open the link on your device or scan the QR code with the Duo Mobile App to add the Duo account and activate vpnuser. Once done, the Duo account is added, and vpnuser is fully activated.

Appendix B. FMC Pre-configuration

After the initial installation, several configuration steps were performed on the FMC to expedite the lab exercises. These configuration steps are detailed in this appendix.

- Configuration A1,1 : Objects
- Configuration A1,2: Demo file policy
- Configuration A1,3: Demo intrusion policy
- Configuration A1,4: Demo SSL policy
- Configuration A1,5: Custom detection list
- Configuration A1,6: Add resetapiuser.
- Configuration A1,7: Install server certificate

Steps

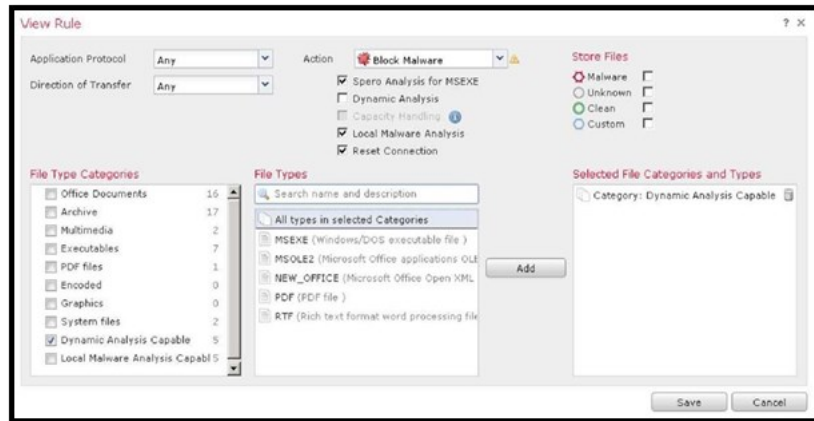
Configuration A1,1: Network Objects

Name	Type	Network
VPNPoolIPs	Range	198.19.10.57-198.19.10.62
LAN_Network	Network	198.19.10.0/24
ISE_Server	Host	198.19.10.130
DNS_Server	Network	198.19.10.100

Configuration A1,2: Demo file policy

16. **Navigate to** Policies > Access Control > Malware & File.
17. **Click** New File Policy. **Enter a name** Demo File Policy. **Click** Save.
18. Click **Add File Rule**. This rule will block malware found in files MSEXE, MSOLE2, NEW_OFFICE and PDFs.
19. **For Action select** Block Malware.
20. Check the Spero and **Local Malware Analysis** checkboxes.
21. Under **File Type Categories**, check *Dynamic Analysis Capable*. **Note** that several file types belong to this category. Click **Add**.

22. Your screen should look like the figure below.



23. Click **Save**. Ignore the warning and click **OK**, when prompted.

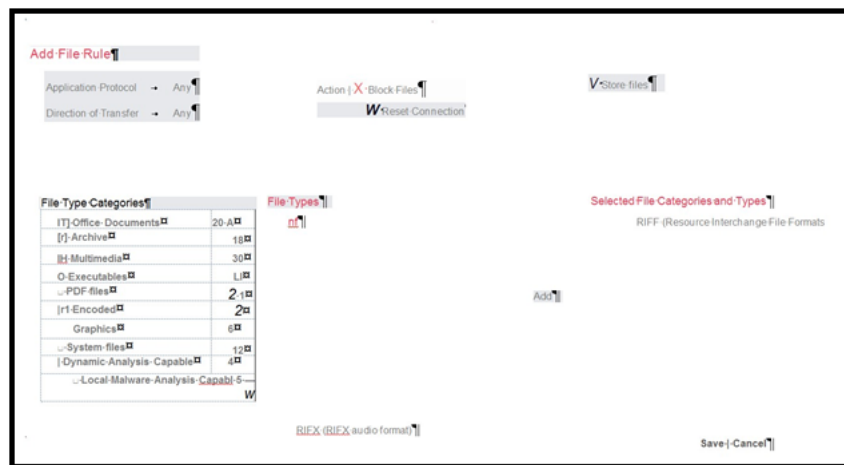
24. Click **Add File Rule**. This rule will block RIFF files. You will use an AVI file to test this rule, since an AVI file is a type of RIFF file. But **Note** that AVI is not listed separately as a file type.

25. **For Action select Block Files.**

26. Under **File Types**, type **riff** into the search box. Select **RIFF** from the list. Click **Add**.

27. Use default values for other settings. Your screen should look like the figure below.

28. Click **Save**.



NOTE: You cannot change the order of the rules you create. The order of the rules does not matter. The action of the rule determines its precedence. The precedence of actions is as follows.

- Block Files
- Block Malware
- Malware Cloud Lookup

- Detect Files

29. Select the **Advanced** tab. Confirm that **Enable Custom Detection List** is selected.
30. Check the **Inspect Archives** checkbox.



NOTE: Archives unable to be inspected are corrupt archive, or archives with a depth that exceeds the Max Archive Depth.

31. Click the **Save** button in the upper-right to save the file policy.

Configuration A1,3: Demo intrusion policy

32. **Navigate to** Objects > Intrusion Rules. **Click** Import Rules.
 - a. **Select the** Rule update or text rule file to upload and install **radio button**.
 - b. **Click Browse** and open the **Snort_Rules.txt** file in the **Files** folder of the Jump desktop.

NOTE: This file contains 2 simple Snort rules that are useful for testing IPS. They do not resemble published snort rules.

```
alert tcp any any -> any any (msg:"ProjectQ replaced"; content:"ProjectQ"; replace:"ProjectR"; sid: 1001001; rev:1;)
alert tcp any any -> any any (msg:"ProjectZ detected"; content:"ProjectZ"; sid: 1001002; rev:1;)
```

The first rule replaces the string ProjectQ with ProjectR. The second detects the string ProjectZ. Since the rules do not specify where the string is in the flow, they could cause issues in a production deployment.

- c. **Click Import**. The import process will take a minute or two. When it completes you will see the **Rule Update Import Log page**. Confirm that 2 rules were successfully imported.
33. **Navigate to** Policies > Access Control > Intrusion.
34. **Click** Create Policy.
 - a. **Set Name to** Demo Intrusion Policy.
 - b. Make sure that **Drop when Inline** is checked.
 - c. Select Balanced Security and Connectivity as **Base Policy**.

Create Intrusion Policy

Policy Information

Name *

Description

Drop when Inline ☒

Base Policy

* Required

d. Click **Create and Edit Policy**.

35. You will now modify the rules states for this new policy.

- a. Click **Rules** under Policy Information menu on the left-hand side of the **Edit Policy** page.
- b. Select **local** from the Category section of the rules. You should see the 2 uploaded rules. The light green arrows on the right of each rule indicate that the rules are disabled for this policy.
- c. Check the checkbox next to the first rule. Select Generate Events from the Rule State drop-down menu. Click OK. Uncheck the checkbox next to the first rule.
- d. Check the checkbox next to the second rule. Select Drop and Generate Events from the Rule State drop-down menu. Click OK.
- e. Clear the filter by clicking on the X on the right side of the Filter text field.
- f. Select **SID** from the **Rule Content** section of the rules. Enter 336 into the **Enter the SID** filter popup. Click **OK**.
- g. Check the checkbox next to the rule. Select **Drop and Generate Events** from the **Rule State** drop-down menu. Click OK.

NOTE: This rule looks for a change to the root home directory in FTP traffic established on port 21. It only looks for traffic coming from the external network, but in our lab, we use the default value of \$EXTERNAL_NET, which is any, so the rule can be triggered in both directions.

An interesting exercise would be to modify this rule to search in FTP traffic in any direction, and to use the appid attribute to detect FTP traffic on any port.

Click **Policy Information** in the menu on the upper left.

Click **Commit Changes**.

Click **OK**.

Configuration A1,4: Demo SSL policy

36. **Navigate to** Objects > Object Management > PKI > Internal CAs.

- a. Click **Import CA**.
 - b. For **Name**, enter Verifraud.
 - c. Click the **Browse** button to the right of the text **Certificate Data or, choose a file**.
 - d. Browse to the **Certificates** folder on the Jump desktop.
 - e. **Upload** Verifraud_CA.cer.
 - f. Click the **Browse** button to the right of the text *Key or, choose a file*.
 - g. **Upload** Verifraud_CA.key.
 - h. Click **Save**.
37. You will exempt from decryption infrastructure devices, such as the FMC and AMP Private Cloud. To do this, create a network object that includes these devices.
 - a. **Navigate to** Objects > Object Management > Network.
 - b. **Click** Add Network > Add Object.
 - c. For **Name**, enter Infrastructure.
 - d. For Network, enter 198.19.10.80-198.19.10.130.
 - e. Click **Save** to save the network object. 3. Navigate to **Policies > Access Control > SSL**.
38. Click the text **Add a new policy** or click the **New Policy** button.
 - a. For **Name**, enter Demo SSL Policy.
 - b. Leave the default action to **Do not decrypt**.
 - c. Click **Save**. Wait a few seconds, and the policy will open for editing.
39. Click **Add Rule**.
 - a. **For Name, enter** Exempt Infrastructure.
 - b. **Leave Action set to** Do Not decrypt.
 - c. **In the Networks tab, under Networks, select** Infrastructure, **and click** Add to Source.
 - d. Click **Add** to add this rule to the SSL policy.
40. Click **Add Rule**.
 - a. **For Name, enter** Decrypt Search Engines.
 - b. **Set Action to** Decrypt - Resign.
 - c. Select **Verifraud** from the drop-down list to the right of the word **with**.
 - d. In the **Applications** tab, under **Application Filters**, search for **Sear**. You will see **Search Engine** under **Categories**. Check this checkbox, and click **Add to Rule**.
 - e. Select the **Logging** tab and check the **Log at End of Connection** checkbox.
 - f. Click **Add** to add this rule to the SSL policy.

41. Click **Add Rule**.
 - a. For Name, enter **Decrypt Other**.
 - b. **Set Action to Decrypt - Resign**.
 - c. Select Verifraud from the drop-down list to the right of the word **with**.
 - d. Select the **Logging** tab and check the **Log at End of Connection** checkbox.
 - e. Click **Add** to add this rule to the SSL policy.
42. Click **Save** to save the SSL policy.

NOTE: The Replace Key checkbox deserves explanation. Whenever the action is set to Decrypt - Resign, Firepower will replace the public key. The Replace Key checkbox determines how the decrypt action is applied to self-signed server certificates.

If Replace Key is deselected, self-signed certificates are treated like any other server certificates. Firepower replaces the key and resigns the certificate. Generally, the endpoint is configured to trust Firepower, and therefore will trust this resigned certificate.

If Replace Key is selected, self-signed certificates are treated differently. Firepower replaces the key, and generates a new self-signed cert. The browser on the endpoint will generate a certificate warning.

In other words, checking the Replace Key checkbox makes the resign action preserve lack-of-trust for self-signed certificates.

Configuration A1,5: Custom detection list

There is a harmless file called Zombies.pdf that will trigger a malware event, assuming the cloud lookup succeeds. Sometimes labs have issues with cloud connectivity. Therefore, this is added to the custom detection list to ensure it will trigger a malware event.

43. **Navigate to** Objects > Object Management > File List.
44. Click the pencil icon to edit the **Custom-Detection-List**.
 - a. Select **Calculate SHA** from the **Add by** drop-down list.
 - b. Click **Browse**.
 - c. Browse to the Files folder on the Jump desktop.
 - d. Select **Zombies.pdf** and click **OK**.
 - e. **Click** Calculate and Add SHAs.
 - f. Click **Save**.

Configuration A1,6: Add restapiuser

It is convenient to have a separate use to use the API Explorer. This allows use of both the FMC and API Explorer at the same time.

45. **Navigate to** System > Users. **Click** Create User.

- a. For **User Name**, enter restapiuser.
- b. For **Password**, enter **C1sco12345** Confirm the password.
- c. Set Maximum Number of Failed Logins to 0.
- d. Check the **Administrator** checkbox.

Configuration A1,7: Install server certificate

By default, the FMC UI uses a self-signed certificate. This is replaced by a certificate signed by the pod AD server, which the Jump PC browsers trust.

46. **Navigate to** Objects > Object Management > PKI > Trusted CAs.
 - a. **Click** Add Trusted CA.
 - b. For **Name**, enter **dCloud**.
 - c. Click the **Browse** button to the right of the text **Certificate Data or, choose a file**.
 - d. Browse to the **Certificates** folder on the Jump desktop.
 - e. **Upload** AD-ROOT-CA-CERT.cer.
 - f. Click **Save**. Connect to the FMC CLI via SSH. Become root by typing **sudo -i**. The Sudo password is **C1sco12345**
 - g. **Type** `cd /etc/ssl and then type cp server* /root.`
 - h. **Type** `cat > /etc/ssl/server.crt`
 - i. From the **Certificates** folder on the Jump desktop edit the file **fmc.cer** with Notepad++.
 - j. Select all, and then copy and paste into the FMC CLI
 - k. Type **Ctrl+D**.
 - l. **Type** `cat > /etc/ssl/server.key`
 - m. From the **Certificates** folder on the Jump desktop edit the file **fmc.key** with Notepad++.
 - n. Select all, and then copy and paste into the FMC CLI
 - o. Type **Ctrl+D**.
 - p. **Type** `pmtool restartbyid httpsd.`

Appendix C. FDM Pre-Configuration

Configuration A1,1: Objects

The following objects have been pre-configured.

Name	Type	Network
Corporate_LAN	Network	198.19.10.0/24
DNS_Server	Host	198.19.10.100
VPN_Pool	Network	198.19.10.64/29
ISE_Server	Host	198.19.10.130
DCloud_LAN1	Network	10.0.0.0/8
DCloud_LAN2	Network	198.19.255.0/24
DCloud_Jumpbox	Host	198.18.133.50

Appendix D. ISE RA VPN Configuration

ISE was configured to support all the lab exercises. In this appendix, this configuration is summarized. Note that there is an ISE link on the Firefox bookmarks toolbar. The credentials should prepopulate. They are username **admin**, password **C1sco12345**

NOTE: This appendix is not a tutorial on ISE. It does not go into details about how ISE is configured. It only covers the details required to configure RA VPN components for the lab exercises in this guide. The configurations are described in a top-down manner. To create this configuration, you would probably prefer to build these objects from the bottom-up.

Authorization policies

47. Navigate to **Policy > Authorization**. The first two policies were created for this lab: **AC-IT-Policy** and **AC-Default-Policy**. These policies reference two authorization profiles: AC-Auth-IT and AC-Auth-Default.

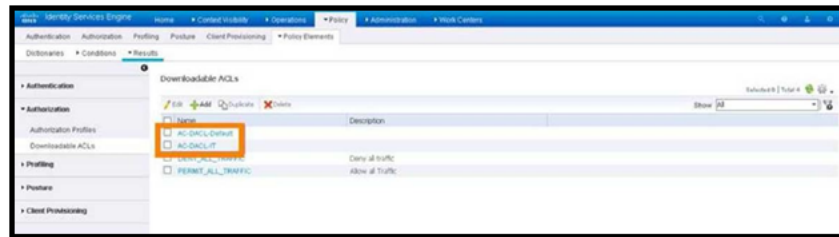
Authorization profiles

48. **Navigate to Policy > Policy Elements > Results > Authorization > Authorization Profiles.** The first two profiles were created for this lab: AC-Auth-Default and AC-Auth-IT.
49. If you drill down into **AC-Auth-Default**, you will see that it references the **DACL AC-DACL-Default**, described below.
50. If you drill down into **AC-Auth-IT**, you will see that it references the **DACL AC-DACL-IT**, described below. It also has two advanced attributes: one for the address pool, and one for the group policy.

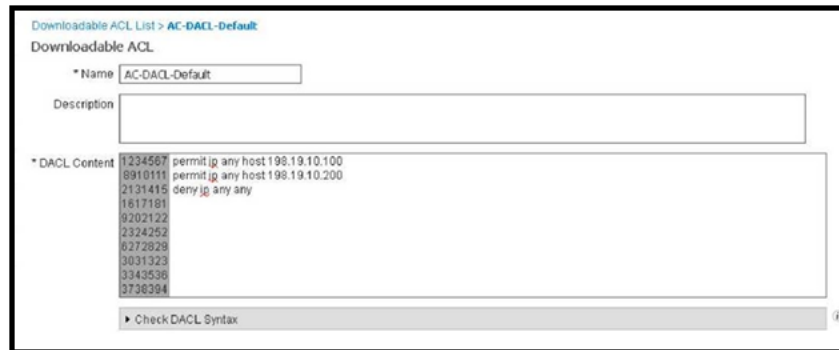


Downloadable ACLs

51. Navigate to **Policy > Policy Elements > Authorization > Downloadable ACLs**. The first two DACLs were created for this lab: **AC-DACL-Default** and AC-DACL-IT.



52. If you drill down into **AC-DACL-Default**, you will see that it restricts access to 198.19.10.100 and 198.19.10.200.



53. If you drill down into **AC-DACL-IT**, you will see that there are no restrictions.

